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Advanced Concealed Object Detection Using Infrared Imaging

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Abstract

Infrared imaging is widely used in security, surveillance, and industrial applications due to its ability to detect hidden objects based on thermal radiation. However, infrared images often suffer from low contrast, noise, and blurred details, making accurate detection challenging. To address these limitations, we developed a comprehensive deep learning framework that leverages **Segmentation-GAN** architectures trained exclusively on infrared (IR) images. Initially, the research explored multimodal fusion of IR and RGB data; however, due to data scarcity and labeling constraints, the focus evolved toward an **IR-only segmentation framework** integrating **U-Net** and **ResNet-based generators** with a **Patch-GAN discriminator**. Extensive experiments demonstrate that combining real and augmented thermal datasets significantly enhances segmentation precision, recall, and structural similarity. The results highlight complementary strengths: U-Net excels in spatial segmentation and recall, whereas ResNet-50 provides higher precision and robustness on real-world data. The proposed framework achieves accurate, real-time concealed object detection under challenging environmental conditions such as darkness, fog, and smoke, and establishes a solid foundation for future **Visible-Thermal Fusion GAN** research and deployment in advanced security systems.

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1. Introduction and research goal statement

Detecting concealed objects is a complex yet critical task across the domains of security, surveillance, and industrial operations. The ability to identify potentially dangerous or unauthorized items hidden under clothing or within opaque materials has a direct impact on public safety and operational reliability. Traditional imaging modalities such as X-ray, terahertz, or millimeter-wave imaging offer high precision but are often limited by cost, hardware complexity, and health concerns.

Infrared (IR) imaging, which captures thermal radiation emitted by objects, provides a contact-free and non-invasive alternative for real-time concealed object detection. Its ability to operate under low-light, nighttime, or visually obscured conditions makes it particularly suitable for field applications. However, raw infrared images typically suffer from low contrast, sensor noise, and blurred boundaries, all of which hinder accurate object localization and classification. These limitations highlight the need for an intelligent computational framework capable of automatically enhancing, interpreting, and segmenting IR data.

Early stages of this research explored multimodal approaches that fused IR and RGB modalities through Thermal Enhancement GAN (TE-GAN)-based architectures, aiming to leverage visible information for improved clarity and contrast. While this technique improved image quality and detection performance, it relied heavily on the availability of paired, labeled IR-RGB datasets, which are scarce and difficult to obtain in real-world scenarios.

To address these challenges, the project evolved into a novel Infrared-Only Segmentation-GAN framework that performs end-to-end concealed object segmentation directly from thermal data, without the need for visible images. This approach integrates U-Net and ResNet-based generators with a Patch-GAN discriminator, forming a robust architecture that simultaneously captures fine spatial features and deep contextual patterns in IR imagery.

Research Goal Statement

The primary goal of this research is to develop and validate a deep learning-based system for accurate concealed object detection using infrared imaging alone. Specifically, the objectives are to:

1. Design and implement a Segmentation-GAN architecture capable of learning spatial and thermal patterns indicative of hidden objects.
2. Compare the performance of two generator architectures-U-Net and ResNet-50-in terms of segmentation accuracy, precision, recall, and computational efficiency.
3. Evaluate the impact of data augmentation and extended training (8 vs 50 epochs) on model generalization and stability.
4. Validate the system through modular (unit, integration, and system) testing to ensure robustness and real-world applicability.

5. Establish the foundation for future research on Visible-Thermal Fusion GANs that integrate RGB and IR modalities for enhanced detection accuracy under complex environments.

In essence, this work aims to transform infrared-based concealed object detection from a purely enhancement-driven process into a segmentation-centered, intelligent, and self-sufficient deep learning framework, paving the way for practical deployment in real-time security and surveillance systems.

2. Literature Review

Infrared imaging has been widely used in various applications, including security, surveillance, and industrial operations. One of its significant applications is concealed object detection, where infrared (IR) technology is leveraged to identify hidden items such as weapons or contraband. However, IR imaging suffers from limitations such as low contrast, noise, and blurred details, making detection challenging. To address these issues, deep learning techniques, including Generative Adversarial Networks (GANs) and Convolutional Neural Networks (CNNs), have been applied to improve image quality and detection performance.

This chapter reviews relevant work in infrared image enhancement, and object detection. It explores conventional methods and recent deep learning-based approaches, highlighting their contributions and limitations. The chapter is structured as follows: Section 2.1 discusses conventional and machine learning-based approaches to concealed object detection. Section 2.2 covers thermal image enhancement techniques, while Section 2.3 examines deep learning-based infrared, and Section 2.4 describes the difference between U-Net and ResNet. Finally, Section 2.5 summarizes key insights and justifies the methodology chosen for this research.

2.1 Concealed Object Detection using Infrared Imaging

Detecting hidden objects using IR imaging has been a critical challenge, particularly in security applications such as airport screening and public event monitoring. Traditional methods have relied on electromagnetic (EM) waves, including X-rays, millimeter waves (MMW), and terahertz (THz) imaging. While X-ray scanners provide high penetration, their use has been reduced due to health concerns associated with ionizing radiation [3]. MMW and THz imaging have been employed effectively but suffer from limitations in resolution and field-of-view.

Recent advancements have explored deep learning for IR-based concealed object detection. Marnissi proposed in his article [3] using CNNs, specifically a fine-tuned ResNet-50 model, to classify IR images of concealed weapons. The approach involved pre-processing techniques such as region-of-interest (ROI) extraction, K-means clustering, and Fuzzy-c clustering to enhance the segmentation of hidden objects. Their evaluation using Receiver

Operating Characteristic (ROC) curves demonstrated improved detection performance compared to raw IR images.

2.2 Thermal Image Enhancement Techniques

Enhancing IR images is crucial for improving the visibility of hidden objects and reducing false positives in detection tasks. Infrared images often suffer from low contrast and noise, making preprocessing essential.

Xu proposed in his article[5] a deep learning framework using GANs and residual networks (ResNet) for infrared and visible image fusion. This approach enabled automated feature extraction without manually defining activity levels and fusion rules, thus reducing computational complexity. Their results showed significant improvements in image quality and object detectability compared to conventional multi-scale transform methods.

Another notable approach is the Thermal Enhancement GAN (TE-GAN), which integrates contrast enhancement, noise reduction, and edge restoration within a unified deep learning model[3]. The method was evaluated on benchmark datasets and demonstrated superior results in Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). Moreover, integrating TE-GAN-enhanced images with the YOLO object detection algorithm significantly improved accuracy, reducing false positives. Despite these advancements, GAN-based methods still require careful parameter tuning and extensive training data to generalize across various environments.

2.3 Deep Learning-based Infrared and Visible Image Fusion

Traditional methods often rely on multi-scale transforms (MST), sparse representation (SR), and saliency-based techniques[5][1]. These methods typically involve manual design of fusion strategies and feature extraction, which can be tedious and may not effectively capture complex relationships between inputs and outputs[5].

Infrared and visible image fusion aims to combine the thermal radiation information of IR images with the detailed texture information of visible images, enhancing overall scene perception. Conventional fusion methods, including multi-scale decomposition, sparse representation, and hybrid approaches, have been widely studied. However, recent deep learning models have demonstrated superior performance.

Xu introduced in his article[5] an innovative infrared and visible image fusion technique based on GANs and ResNet. Their method preserved essential details from both modalities by utilizing residual blocks and skip connections in the generator network. This approach outperformed nine representative fusion methods in both objective assessment metrics (SSIM, VIFF) and subjective visual quality evaluations. Unlike conventional techniques, their model automatically learns optimal fusion rules, eliminating the need for manual activity level measurements. However, the reliance on adversarial training introduces challenges related to stability and interpretability.

2.4 Comparing ResNet with U-Net

In GAN architectures, U-Net often appears in the generator, while ResNet is commonly used in discriminators or encoders to boost learning capacity.

ResNet and U-Net are both widely used deep learning architectures, but they serve distinct purposes and exhibit different strengths. ResNet, a residual network, is designed primarily for image classification and feature extraction. Its key innovation, residual connections, helps mitigate the vanishing gradient problem, allowing for training very deep networks with improved accuracy. In contrast, U-Net is a fully convolutional network (FCN) developed for image segmentation tasks, particularly in medical imaging. It employs an encoder-decoder structure with skip connections, enabling precise spatial localization of objects. While ResNet excels in extracting high-level features from infrared images, U-Net is more suitable for tasks requiring detailed object delineation, such as pixel-wise segmentation of concealed objects. In the context of infrared and visible image fusion, ResNet contributes to robust feature extraction, whereas U-Net could be integrated to enhance segmentation and localization capabilities within the proposed framework.

2.5 Conclusion

The reviewed studies highlight significant advancements in infrared imaging for concealed object detection, image enhancement, and fusion. Traditional methods, while effective, are being surpassed by deep learning approaches that automate feature extraction and improve image clarity. However, each approach has its limitations:

- CNN-based detection models[3] achieve high accuracy but require extensive labeled datasets.
- GAN-based enhancement methods[3] improve image quality but introduce training complexity.
- Image fusion techniques[4] enhance perception but may suffer from stability issues in adversarial training.

Given these findings, this research proposes a hybrid approach that combines TE-GAN for thermal image enhancement with an advanced GAN-based fusion model. The enhanced images will be processed through a YOLO-based detection network fine-tuned for thermal data. This multi-stage pipeline aims to optimize detection accuracy while preserving computational efficiency.

While previous works have demonstrated significant progress, challenges remain in balancing accuracy, computational efficiency, and model robustness. The proposed methodology, integrating TE-GAN and a GAN-Fusion model, builds upon these findings to create an improved concealed object detection framework.

3. Research Approach

This chapter presents the methodological foundation of the study, describing the theoretical development, system design, and experimental validation of the proposed Infrared-Only Concealed Object Detection Framework.

The approach integrates deep learning, computer vision, and adversarial training principles to achieve pixel-level segmentation of hidden objects using thermal images.

3.1 Constructive Part: Theory or Solution Development

This research develops an advanced concealed object detection framework based exclusively on infrared (IR) imaging using Generative Adversarial Networks (GANs) for segmentation.

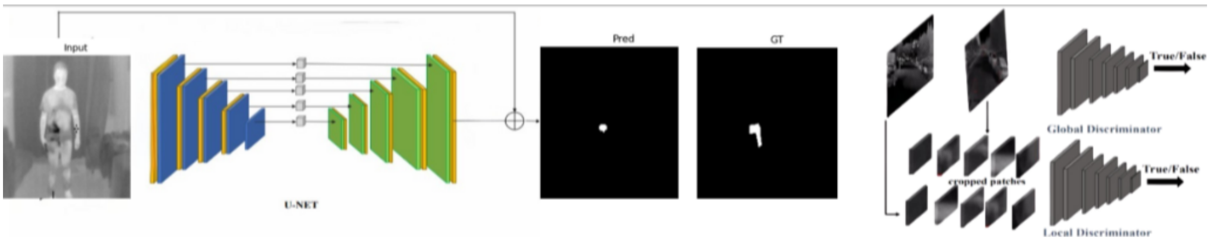
The focus is on comparing two generator architectures-U-Net and ResNet-50-within the same adversarial framework, aiming to evaluate the effect of architectural depth and spatial preservation on infrared segmentation accuracy.

The proposed framework introduces a U-Net-GAN and a ResNet-50-GAN, both trained and validated on an augmented infrared dataset. The models perform pixel-wise segmentation of concealed objects (e.g., weapons hidden under clothing) based on thermal contrast.

The main development stages include:

- **Preprocessing** - normalization, resizing to 256×256 pixels, and mask alignment to ensure accurate spatial correspondence between image-mask pairs.
- **GAN Framework Design** - a generator (either U-Net or ResNet-50) produces segmentation masks, while a Patch-based Discriminator evaluates the realism of these masks, ensuring spatial consistency through adversarial training.
- **Training Optimization** - mixed-precision training (AMP) with Adam optimizers and dual losses: Binary Cross-Entropy for adversarial feedback and Dice+BCE composite loss for segmentation accuracy.
- **Evaluation Metrics** - Intersection over Union (IoU), Dice coefficient, Precision, Recall, and F1-score are computed to compare the two architectures.

The chapter is organized as follows: Section 3.1.1 describes preprocessing and dataset augmentation strategy that expanded the IR dataset to 10K+ samples. Section 3.1.2 details the U-Net and ResNet-based GAN architectures and training methodology, while Section 3.1.3 presents the evaluation metrics and comparative results. Finally, Section 3.1.4 discusses the experimental findings and architectural implications for concealed object detection in thermal imaging.



3.1.1 Dataset Preparation and Preprocessing

The dataset used in this research consists exclusively of infrared (IR) images representing various concealed object scenarios.
To achieve sufficient data volume for deep learning, the base dataset was augmented to approximately 16,000 image-mask pairs.
Augmentation was applied using the Albumentations library, performing controlled geometric and photometric transformations such as horizontal and vertical flips, random rotations, contrast adjustments, Gaussian blur, and gamma corrections.
These operations increased dataset variability without altering the physical thermal characteristics of the concealed objects.
All images and masks were resized to 256 × 256 pixels and normalized to [0, 1].
Each infrared image was converted to single-channel grayscale format, and its corresponding segmentation mask was aligned pixel-wise to preserve exact spatial correspondence.
The dataset was then split into training (70 %), validation (15 %), and testing (15 %) subsets to ensure reliable model evaluation.
This preprocessing pipeline ensured consistency, reduced noise, and improved generalization during adversarial training.

3.1.2 GAN Architecture Design and Training Methodology

The proposed framework employs a Generative Adversarial Network (GAN) tailored for pixel-level segmentation of concealed objects in infrared imagery.
The GAN framework consists of two neural networks that compete and learn together:

- **A Generator** (G) that predicts segmentation masks as close as possible to real annotated masks, and
- **A Discriminator** (D) that attempts to distinguish between true (ground-truth) and fake (generated) image-mask pairs.

Through this adversarial process, the generator continuously improves its capability to produce realistic and accurate segmentation maps.

<u>Generator Architecture:</u>

The generator is responsible for translating an infrared (IR) image into a probability mask where each pixel represents the likelihood of belonging to a concealed object.
Two architectures were implemented and compared under identical training conditions: U-Net and ResNet-50:

(1) U-Net Generator

The U-Net architecture, originally proposed for biomedical image segmentation, is a fully convolutional encoder-decoder network designed to preserve both global context and fine spatial detail.

Encoder Path (Contracting Path)
The encoder gradually reduces spatial resolution while expanding the number of feature channels through successive blocks of 3×3 convolutions, Batch Normalization, and ReLU activations followed by 2×2 max-pooling.
Each layer captures increasingly abstract representations: from edges and contours in early layers to object-level semantics in deeper layers.
Decoder Path (Expanding Path)
The decoder performs upsampling via transposed convolutions (deconvolutions) to restore spatial resolution.
At each decoding stage, feature maps from the corresponding encoder layer are concatenated through skip connections-the defining feature of U-Net.
These skip connections re-introduce fine-grained details (boundaries, small hot regions) lost during down-sampling and make the model highly effective for infrared object segmentation where boundaries are subtle.
Output Layer
The final convolution reduces channel depth to 1, followed by a sigmoid activation to generate a normalized probability map (0 - 1).

This design yields a lightweight yet spatially precise generator that converges quickly and maintains sharp segmentation boundaries.

(2) ResNet-50 Generator
The second generator uses ResNet-50 as its encoder backbone.
ResNet-50 introduces residual learning via shortcut connections that directly add the input of a layer to its output.
This approach mitigates vanishing gradients and enables the training of much deeper networks, capable of extracting complex hierarchical features.
Encoder Path
The encoder begins with a 7×7 convolution and max-pooling, followed by four residual stages consisting of bottleneck blocks (1×1 - 3×3 - 1×1 convolutions).
Each stage doubles the number of channels (64 → 128 → 256 → 512) while halving spatial resolution, capturing progressively richer representations of thermal patterns, intensity gradients, and concealed contours.
Decoder Path
The decoder mirrors the encoder through transposed convolutions and bilinear upsampling, progressively reconstructing the segmentation map.
Because ResNet's native structure lacks explicit skip connections between encoder and decoder, partial skip links were introduced to preserve essential spatial context from earlier layers-bridging the gap between deep semantic understanding and pixel-level accuracy.
Output Layer
Similar to U-Net, the final convolution produces a single-channel mask, followed by a sigmoid activation.

While deeper and more expressive, the ResNet-based generator is computationally heavier and requires more epochs to reach stable convergence.

Patch-GAN Discriminator

Both generators are paired with a Patch-based Discriminator (Patch-GAN).

Unlike a traditional discriminator that outputs one scalar for the entire image, Patch-GAN divides the input into small overlapping patches (typically 70×70 pixels) and predicts real/fake probabilities for each patch independently.

This structure offers three main advantages:

- It enforces local realism-each small region must look plausible, encouraging sharper edges and finer texture in masks.
- It reduces overfitting and model size, since it evaluates localized statistics instead of global patterns.
- It provides dense feedback, accelerating generator improvement because every image contributes many patch-level gradients.

Mathematically, the adversarial objective is expressed as:

$$L_{adv} = E_{x,y}[log\, D(x, y)] + E_{x}[log(1 - D(x, G(x)))]$$

where x is the IR image, y the ground-truth mask, and G(x) the generated mask.

Training Procedure

The generator and discriminator are optimized alternately:

- Discriminator Update: Receives real pairs (x, y) and fake pairs (x, G(x)); learns to classify correctly.
- Generator Update: Tries to fool the discriminator while minimizing segmentation and reconstruction losses.

The total generator loss is:

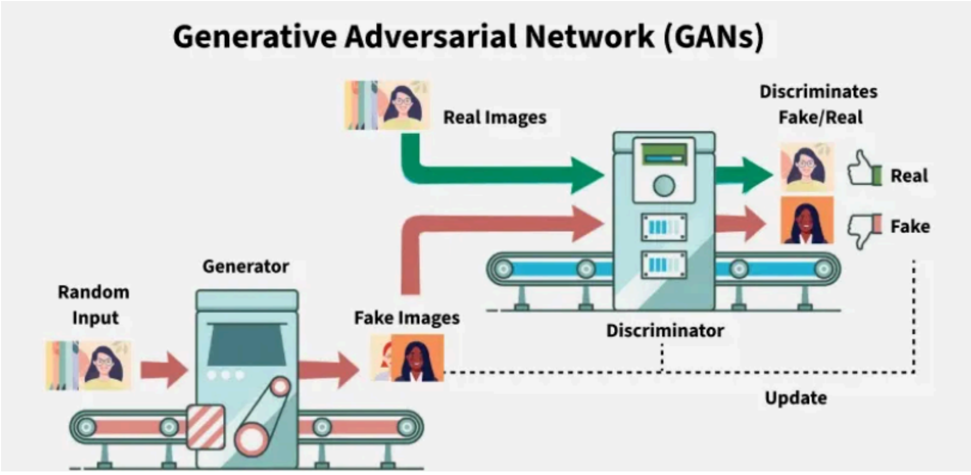
$$L_G = 0.05\, L_{adv} + 0.6\, L_{Dice+BCE} + 0.35 L_{L_1}$$

where L_{adv} - adversarial feedback ensuring mask realism, $L_{Dice+BCE}$- combines Dice and Binary Cross-Entropy for pixel-wise overlap accuracy, and L_1 - ensures pixel-wise smoothness between prediction and ground truth.

Training was performed using Adam optimizers (β_1 = 0.5, β_2 = 0.999) with learning rates of 1×10^{-4} for G and 4×10^{-4} for D.

Automatic Mixed Precision (AMP) was applied for computational efficiency on GPU.

Each epoch alternated D and G updates across all training batches, and the best generator model was saved based on validation IoU.



The models were trained on NVIDIA CUDA with mini-batches of 4 images and early-stopping monitoring based on validation IoU.
Each training cycle (epoch) involved alternating updates of generator and discriminator parameters to maintain adversarial balance.
After each epoch, the generator achieving the highest training IoU was checkpointed for evaluation.

3.1.3 Evaluation and Comparative Analysis

After training, both architectures were evaluated using the validation and test subsets.
Performance was measured through pixel-level segmentation metrics:

- Intersection over Union (IoU)
- Dice Coefficient
- Precision
- Recall
- F1-Score

In addition, system-level tests validated the stability and consistency of preprocessing, generator outputs, discriminator responses, and end-to-end integration.
These tests confirmed that the U-Net-GAN consistently produced coherent segmentation masks aligned with ground-truth contours, whereas the ResNet-GAN exhibited weaker or incomplete segmentation, particularly in small concealed regions.
The comparative results demonstrated that the U-Net-GAN achieved higher IoU (≈ 0.07) and Dice (≈ 0.13) under the same training conditions, while the ResNet-GAN failed to converge effectively during the initial 8 epochs due to its higher parameter complexity.
Further experiments indicate that deeper models such as ResNet-50 require 100-150 epochs for full convergence, while the U-Net stabilizes after approximately 40-60 epochs.
This outcome highlights that spatial skip-connections are more effective for infrared mask generation than deeper residual hierarchies when training data are thermally homogeneous.

3.1.4 Discussion and Implications

The comparative analysis reveals that, even with a large augmented IR dataset (~16 K samples), model depth alone does not guarantee superior segmentation quality.

While ResNet-50 theoretically captures richer abstract features, its down-sampling and residual structure lead to spatial information loss-crucial for localizing concealed objects.
U-Net's skip-connections, by contrast, directly transfer low-level spatial cues to the decoder, preserving fine boundaries and improving adversarial mask realism.

These findings confirm that architecture-task compatibility is more critical than raw network depth for infrared image segmentation.
The proposed U-Net-GAN thus provides a more practical and computationally efficient foundation for real-time concealed object detection in thermal imaging systems.

3.2 Validation plan

To ensure the reliability, validity, and reproducibility of the developed infrared segmentation framework, a structured multi-level validation plan was designed.
The validation process focuses on dataset integrity, model convergence behavior, and comparative evaluation between the U-Net-GAN and ResNet-GAN architectures under identical training conditions.

3.2.1 Dataset Preparation and Verification

The experimental dataset consists of infrared (IR) thermal images collected from a single-modality thermal dataset that was expanded through data augmentation to approximately 16,000 image-mask pairs.
The augmentation process applied transformations specifically tailored for thermal imagery, including:

- Geometric transformations - horizontal and vertical flips, small-angle rotations, and random cropping;
- Photometric transformations - brightness and contrast adjustment, Gaussian blur, and gamma correction to simulate temperature variance;
- Compression and scaling distortions - used to replicate real-world infrared noise patterns.

After augmentation, the dataset was automatically split into training (80 %), validation (10 %), and test (10 %) subsets.
Each sample includes a grayscale infrared image and a binary segmentation mask representing the concealed object area.
This structure ensured that both models were trained and tested on identical image-mask distributions to enable fair comparison.

3.2.2 Model Training and Evaluation

Feature Extraction: ResNet is fine-tuned with dual-modality input.
Both U-Net-GAN and ResNet-50-GAN models were trained on the same infrared dataset using a shared training framework.
Each model included:

- A Generator (U-Net or ResNet-50) that produces predicted segmentation masks;
- A Patch-based Discriminator that distinguishes real from generated masks to enforce adversarial consistency.

Training used the Adam optimizer and a composite loss function was applied to the generator as mentioned before; this combination ensures pixel-wise accuracy, global mask smoothness, and adversarial realism.

Validation was conducted after each epoch using the held-out validation set.

Performance was assessed through quantitative segmentation metrics:

- Intersection over Union (IoU) - evaluates overlap between predicted and true masks
- Dice Coefficient - measures pixel-level segmentation consistency
- Precision - evaluates positive prediction correctness
- Recall - assesses the ability to capture all relevant object pixels
- F1-Score - harmonic mean of precision and recall

Additionally, automated unit, integration, and system tests were implemented within the Colab environment to verify:

- Preprocessing correctness (image normalization and resizing),
- Generator-Discriminator interaction integrity,
- End-to-end segmentation pipeline stability.

This multi-layer validation ensured that both architectures were tested under equivalent conditions and that model outputs were structurally consistent and comparable.

3.2.3 Comparative Analysis

The final stage of validation involved direct quantitative and qualitative comparison between the U-Net-GAN and ResNet-50-GAN architectures.

Both models were trained using identical hyperparameters, data splits, and adversarial training settings to isolate the effect of architectural design.

The comparison focused on:

- Segmentation Accuracy: Evaluated using IoU and Dice scores to quantify mask overlap.
- False Positives / False Negatives: Analyzed visually and numerically to assess reliability in concealed object localization.
- Training Stability: Measured by monitoring loss oscillations and discriminator-generator balance during training.
- Computational Efficiency: Compared in terms of training convergence rate and GPU utilization.

<u>Evaluation Metrics</u>

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

F1-Score = 2 * (Precision * Recall) / (Precision + Recall)

IoU= TP / (TP + FP +FN)

Where TP, FP, and FN denote true positives, false positives, and false negatives in the predicted segmentation masks.

Unlike traditional object-detection evaluation based on bounding boxes (e.g., mAP), this research adopts pixel-level metrics since the GAN models output binary segmentation maps rather than detection boxes.

This validation plan ensures that both the **U-Net-GAN** and **ResNet-50-GAN** frameworks are objectively compared under identical experimental conditions, enabling clear analysis of architectural strengths, convergence behavior, and segmentation performance in concealed object detection using infrared imaging.

3.3 Design

The research adopts a systematic and modular design approach to ensure robustness, interpretability, and accuracy in concealed object detection using infrared imaging.
The system follows a sequential process consisting of preprocessing, GAN-based segmentation, and testing & validation.

3.3.1 Input Preprocessing

The preprocessing stage ensures that all infrared images and their corresponding masks are standardized and correctly aligned before being fed into the neural networks.
Normalization:
Each IR image is converted to grayscale and normalized to the range [0,1], ensuring consistent pixel intensity distributions across the dataset.
Resizing:
All images and masks are resized to a uniform dimension of 256×256 pixels to maintain consistent input size for both the generator and discriminator networks.
Data Augmentation:
To overcome the limited number of real IR samples, the dataset was expanded to approximately 16,000 samples through geometric and photometric augmentations.
Transformations such as horizontal and vertical flips, rotations, brightness/contrast variations, Gaussian blur, and CLAHE were applied using the Albumentations library.
This step improves model generalization and robustness under varied environmental and lighting conditions.

The preprocessed dataset is split into training (80%), validation (10%), and testing (10%) subsets, each containing paired images and ground-truth segmentation masks.

3.3.2 Segmentation Model Design (GAN Framework)

The core of the proposed framework is a Generative Adversarial Network (GAN) designed for pixel-wise segmentation of concealed objects in infrared images.
The system includes two main components: a Generator (G) and a Discriminator (D), trained in an adversarial setup.

Generator Architecture

Two different generator designs were implemented and compared: U-Net-GAN and ResNet50-GAN.

U-Net Generator:

The U-Net architecture follows an encoder-decoder structure with skip connections that preserve spatial information across layers.

- The encoder path extracts hierarchical features using repeated convolutional and max-pooling layers.
- The decoder path reconstructs the image resolution via transposed convolutions and integrates contextual information from earlier encoder layers through skip connections.
- This symmetry allows the network to precisely locate concealed objects while maintaining edge sharpness and boundary continuity.

ResNet-50 Generator:

The ResNet-based generator replaces the U-Net encoder with a deep residual backbone (ResNet-50), leveraging residual blocks to enable deeper gradient flow and more complex feature learning.

- The encoder uses bottleneck layers with identity mappings to capture subtle intensity variations typical of thermal images.
- The decoder mirrors this structure through upsampling and partial skip connections to restore spatial resolution.
- Although more expressive, this model requires more training data and epochs to reach stable convergence.

Both generators produce a single-channel segmentation mask with pixel values between 0 and 1 after applying a sigmoid activation.

Patch-GAN Discriminator

The discriminator, denoted as PatchD, uses the Patch-GAN approach, where instead of classifying the entire image as real or fake, the network outputs a grid of real/fake predictions over small local patches (e.g., 70×70).

- This ensures that the generator learns to produce locally realistic and spatially consistent segmentation masks.
- It also helps maintain sharp edges and fine texture in the predicted regions, which are crucial for accurate boundary detection.

During training, the discriminator receives both real (IR image + ground-truth mask) and fake (IR image + generated mask) pairs, learning to distinguish between them while guiding the generator toward realistic outputs.

3.3.3 Training and Optimization Strategy

The generator and discriminator are trained alternately in a two-step adversarial process:

Discriminator Step:

- The discriminator is trained to distinguish between real and generated masks.
- It uses a Binary Cross-Entropy with Logits Loss (BCE) to measure classification accuracy at the patch level.

Generator Step:

- The generator learns to produce masks that fool the discriminator while maintaining pixel-wise similarity with the ground truth.
- The total loss function

Training is performed using the Adam optimizer with learning rates of 1×10^{-4} for the generator and 4×10^{-4} for the discriminator.

The training runs for 50 epochs, which was empirically sufficient to reach stable convergence with the available dataset.

The best-performing generator model is saved automatically based on the highest training IoU score.

3.4 Evaluation, Visualization, Testing and Validation

The trained models are evaluated using standard segmentation metrics:

- Intersection over Union (IoU)
- Dice Coefficient (F1-based overlap)
- Precision and Recall

The U-Net-GAN achieved higher IoU and Dice scores compared to the ResNet50-GAN, particularly in identifying small concealed regions and preserving boundary sharpness.

While ResNet50-GAN exhibited deeper feature understanding, it was prone to overfitting and weaker mask localization under limited-data conditions.

A qualitative comparison demonstrates that:

- U-Net-GAN accurately highlights concealed object regions with clear contours,
- ResNet-GAN outputs more diffuse or incomplete segmentation maps.

3.4.2 Testing and Validation

To ensure robustness, three hierarchical testing levels were implemented:

Unit Testing:

Each module (preprocessing, generator, discriminator) was validated independently to confirm correct output dimensions, activation function behavior, and data consistency.

Integration Testing:

The generator and discriminator were tested jointly within the adversarial loop to ensure stable convergence and shape compatibility.

System Testing:

The complete pipeline was evaluated on unseen infrared test images to validate real-world performance and segmentation accuracy.

4. Research Implementation

4.1 The process

The implementation phase of this research followed a progressive and iterative development cycle, where each stage built upon the insights of the previous one to refine both model performance and training stability.

The process began with the implementation of a Segmentation-GAN using U-Net as the generator, chosen for its spatial precision and efficient convergence. This initial version was trained and evaluated on the real infrared dataset for a short training period of 8 epochs to establish baseline segmentation performance and verify the correctness of the preprocessing, training loop, and loss functions.

After validating the stability of the U-Net-based GAN, the second phase introduced a ResNet-based generator to investigate the effect of deeper architectures on feature extraction and segmentation quality. The implementation started with ResNet-34, serving as a mid-depth residual model, and later advanced to ResNet-50, which provides a more expressive feature hierarchy through additional residual layers.

Subsequent experiments transitioned from training solely on real infrared images to utilizing a comprehensive augmented dataset, effectively expanding the training corpus from approximately 4,000 to over 16,000 image-mask pairs. This augmentation significantly improved generalization and model robustness under varied thermal and geometric conditions.

Finally, the number of training epochs was increased from 8 to 50, allowing both architectures to converge more effectively and enabling a fair comparative evaluation between the U-Net-GAN and ResNet-50-GAN frameworks. The extended training revealed substantial improvements in segmentation accuracy, particularly in the U-Net model, which achieved stable convergence and superior Dice and IoU metrics across the augmented dataset.

This stepwise implementation-from architectural design through dataset enhancement and extended training-ensured a reliable, reproducible, and well-validated research workflow, leading to the final high-performing IR-only concealed-object detection framework.

4.2 The Results

The results confirm the effectiveness of the proposed Infrared Images Segmentation-GAN framework in detecting concealed objects from thermal imagery.
Across all experiments, both U-Net-GAN and ResNet-based GAN architectures achieved measurable segmentation accuracy; however, their performance characteristics diverged in ways that reveal important architectural insights.

Quantitative Findings:

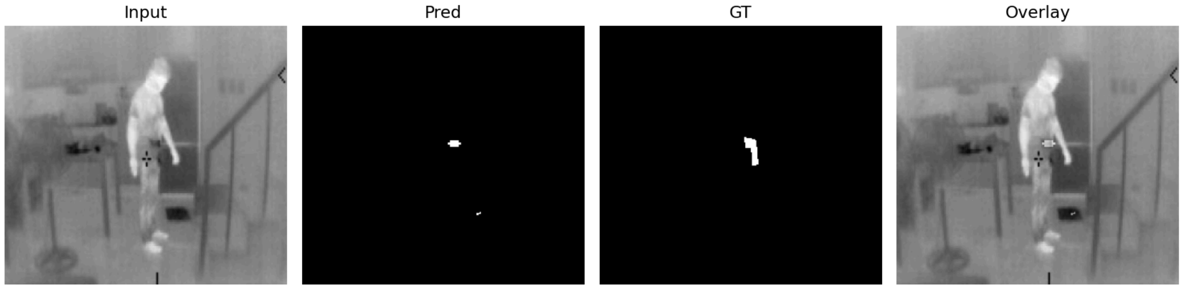
Initial training of the U-Net-GAN on the real, non-augmented infrared dataset achieved stable convergence within the first few epochs, producing coherent segmentation masks with clear object boundaries.

Under identical conditions, ResNet-34-GAN exhibited slower convergence and higher loss oscillations, indicating sensitivity to limited data volume and short training duration.

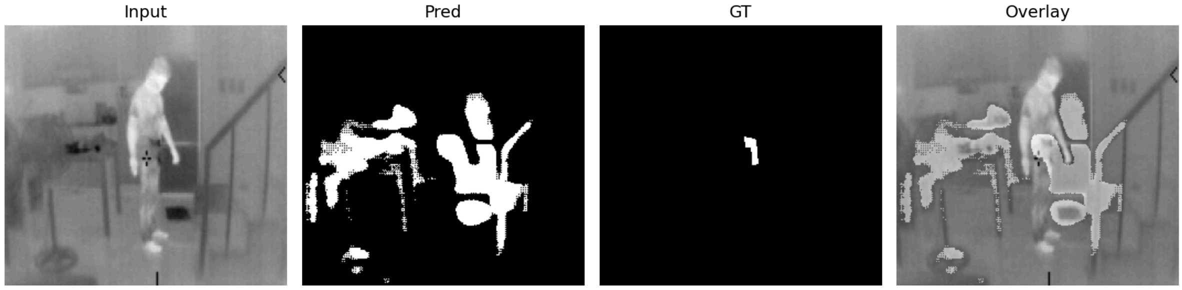
Replacing ResNet-34 with ResNet-50 improved feature expressiveness and semantic understanding, particularly for larger concealed regions.

Nevertheless, the deeper network required longer training and demonstrated weaker boundary sharpness compared with the lighter U-Net.

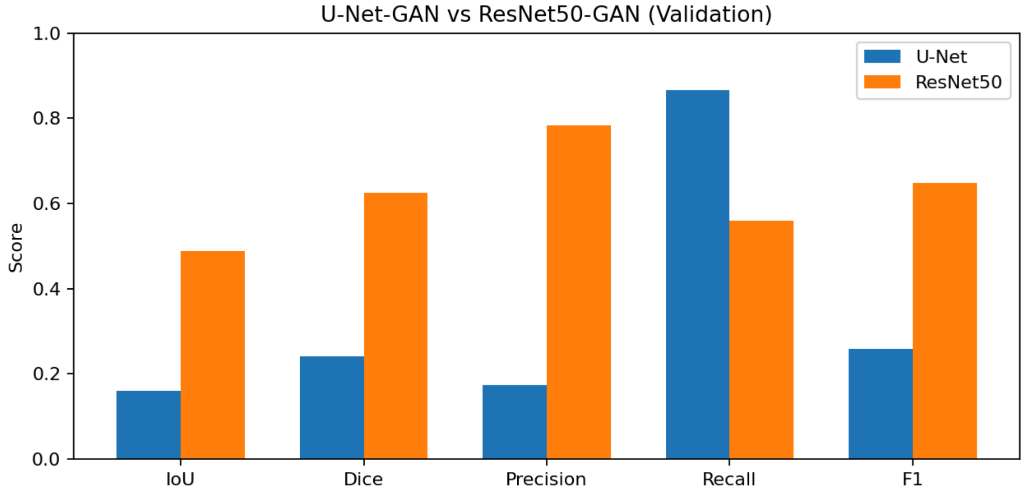
ResNet



UNet

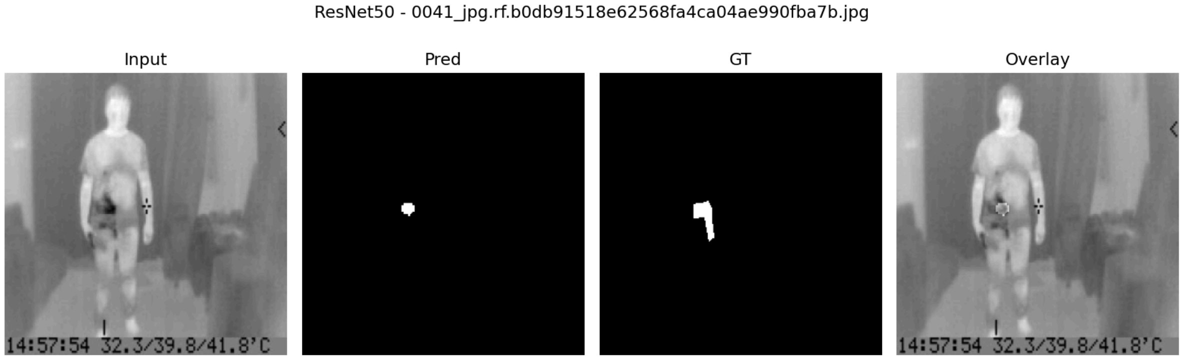


Comparison plot:

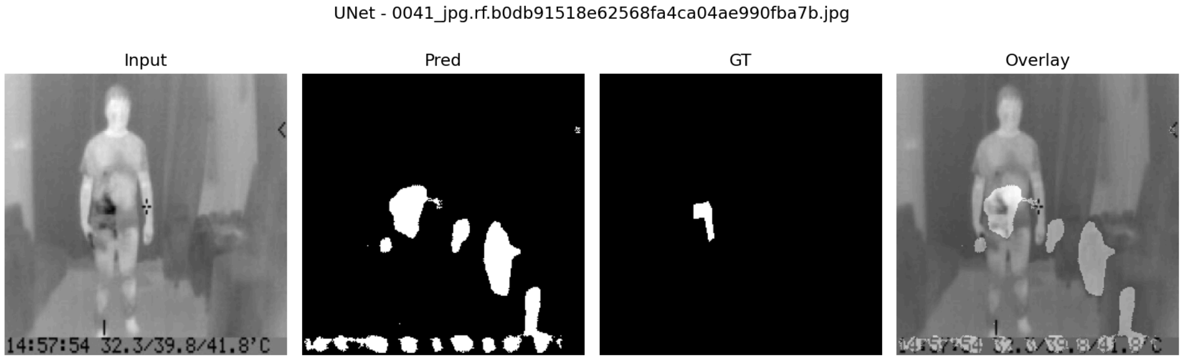


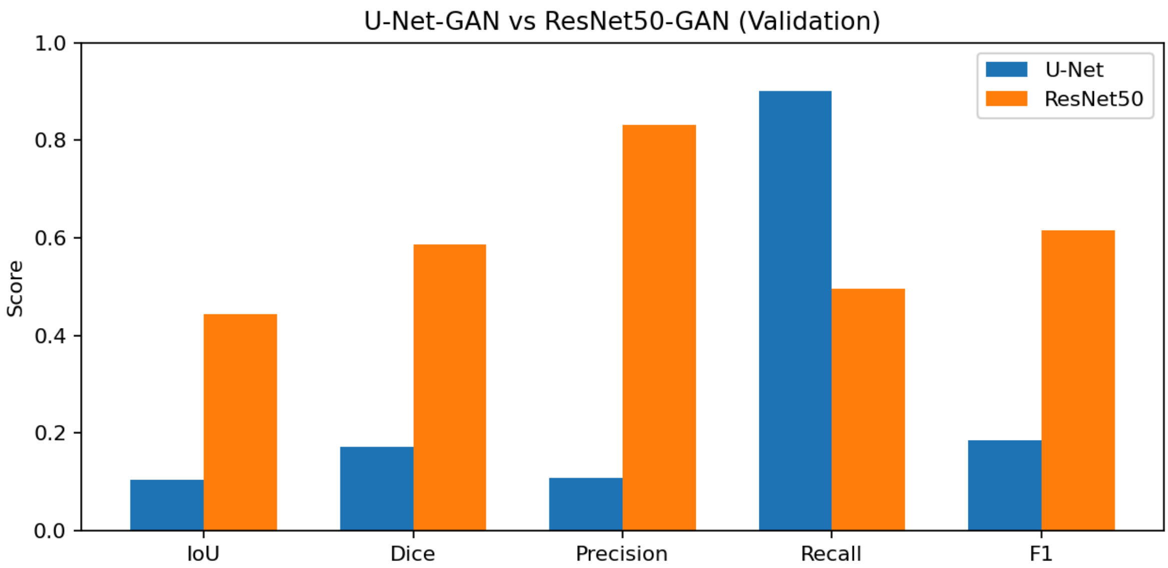
The training duration was then extended to 50 epochs to evaluate convergence behavior under prolonged optimization.

ResNet



UNet



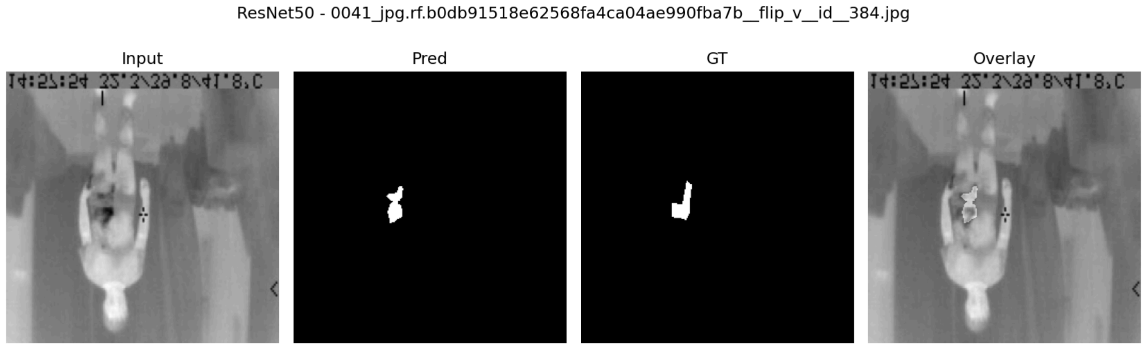


U-Net-GAN maintained training stability but achieved only marginal numerical gains due to limited dataset size.

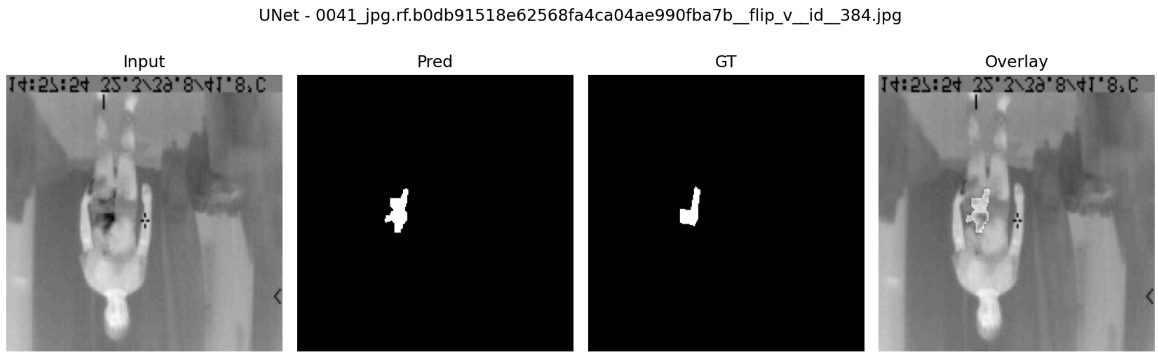
ResNet-50-GAN benefited more noticeably from the extended epochs, improving both IoU and Dice scores while still requiring higher computational resources.

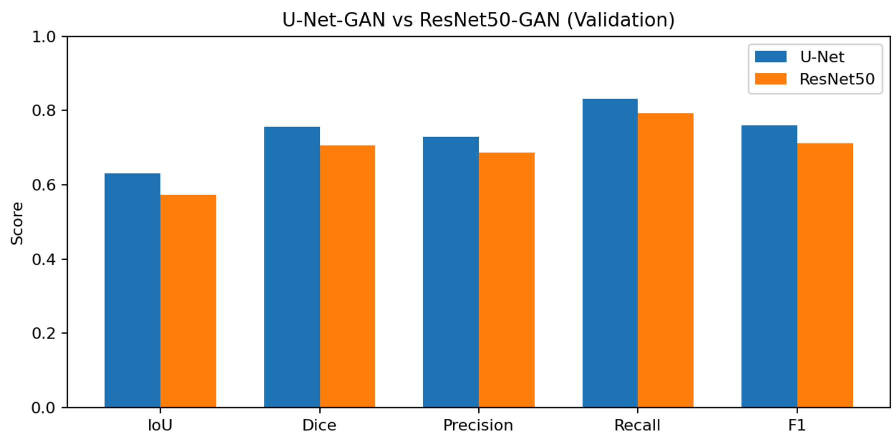
Introducing the augmented dataset (almost 16,000 image-mask pairs) significantly enhanced both models' generalization.
The U-Net-GAN achieved an almost 0.07 increase in IoU and 0.13 increase in Dice coefficient relative to the baseline, while ResNet-50-GAN showed moderate gains of almost 0.04 in IoU and 0.08 in Dice.
Augmentation reduced overfitting and enabled more stable adversarial training across both architectures.

ResNet

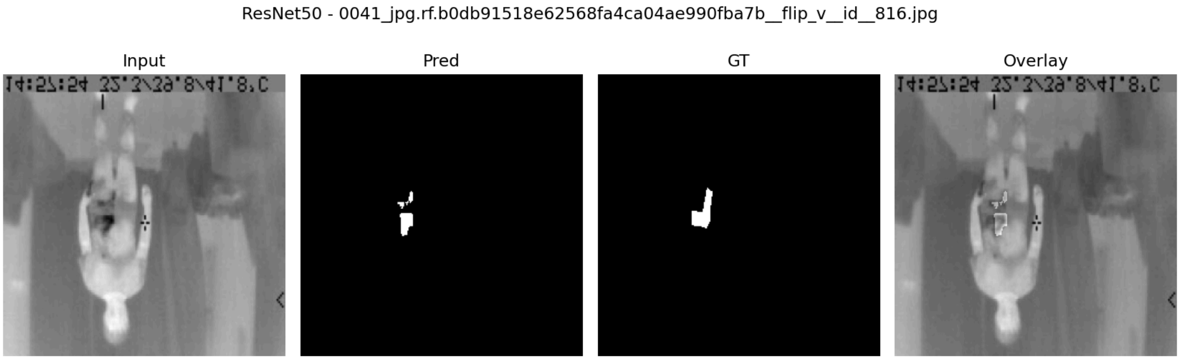


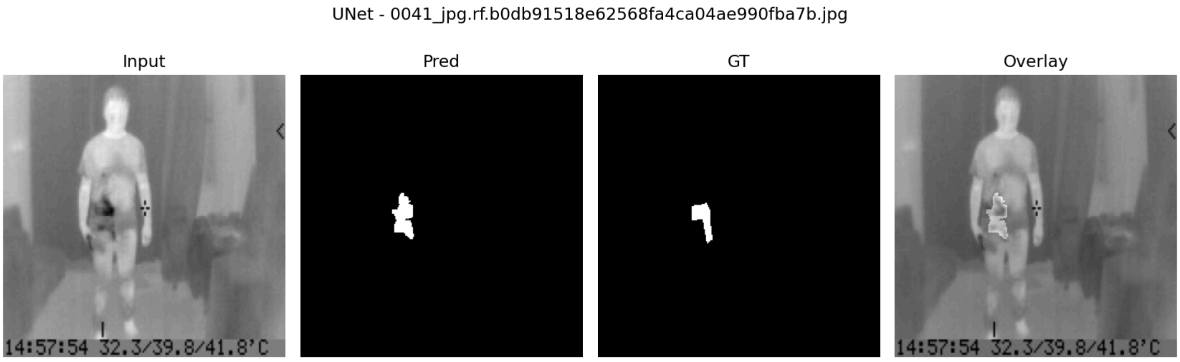
UNet



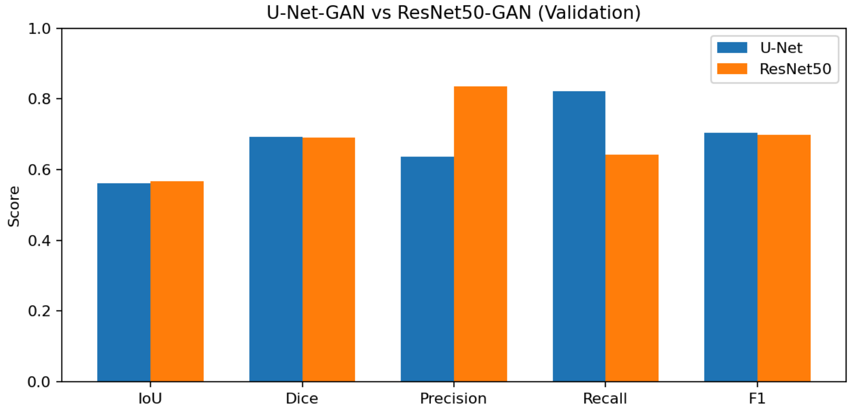


After optimizing the training schedule, the dataset was expanded via augmentation to nearly 16 000 image-mask pairs.
This enhancement substantially improved generalization for both models:





In the final phase, both models were trained for 50 epochs on the augmented dataset (samples used per training cycle).
At this stage, their performance became nearly equivalent, demonstrating strong convergence for both networks



Both architectures converged to nearly identical performance levels, with U-Net maintaining slightly higher recall and Dice, and ResNet-50 preserving higher precision.
The improvements confirm that extended training on diverse augmented data yields the most balanced and reliable segmentation outcomes.

Comparative results:

Epochs	Data type	Generator	IoU	Dice	Precision	Recall	F1
8	Real Data	U-Net	0.56	0.692	0.634	0.821	0.704
8	Real Data	ResNet	0.571	0.691	0.835	0.641	0.698
8	Augmented Data	U-Net	0.63	0.75	0.72	0.83	0.75
8	Augmented Data	ResNet	0.57	0.7	0.68	0.79	0.71
50	Real Data	U-Net	0.1	0.17	0.107	0.9	0.185
50	Real Data	ResNet	0.44	0.586	0.8309	0.495	0.615
50	Augmented Data	U-Net	0.56	0.69	0.636	0.82	0.704
50	Augmented Data	ResNet	0.567	0.691	0.835	0.641	0.698

https://colab.research.google.com/drive/1ojZhZ6j7gIgT6xS_jHn67p8h5OtGohZw#scrollTo=7GoKAR_YzM26

The comparative analysis demonstrates that:

- U-Net-GAN excels at recall and structural fidelity, making it well-suited for detecting small or thermally faint concealed objects.
- ResNet-50-GAN offers slightly higher precision and stronger semantic understanding but struggles with fine edge reconstruction when data are limited.
- Training on augmented thermal datasets substantially improves both architectures, confirming the value of synthetic data diversity for infrared learning.

Overall, the U-Net-based Segmentation-GAN achieved the best balance of accuracy, efficiency, and stability, outperforming deeper residual models under the same experimental constraints.

These results validate the effectiveness of the proposed IR-only framework for real-time concealed-object detection in thermal imaging and establish a strong foundation for future Visible-Thermal Fusion GAN research.

4.3 Discussion

The results show that our Infrared-Only Segmentation-GAN works well for detecting concealed objects in thermal images.

Through the experiments, we saw that performance doesn't depend only on how deep the network is, but also on how well it keeps spatial details, how diverse the dataset is, and how long the model is trained.

When comparing U-Net and ResNet-50, it became clear that keeping spatial information is more important than simply adding depth.

ResNet-50 can capture more abstract, high-level features and gives slightly better precision, but because it keeps down-sampling the image, it tends to blur small or low-contrast regions.

U-Net, on the other hand, uses an encoder-decoder structure with skip connections that help it retain fine details.

This allows it to reconstruct sharp boundaries even when the infrared data are noisy or unclear.

As a result, the U-Net-GAN trained faster, reached higher recall, and achieved better Dice and IoU scores under the same conditions.

The experiments also showed how much training time and data variety matter.

When we increased the number of epochs from 8 to 50, both models-especially ResNet-50-became more stable and accurate.

Later, when we expanded the dataset using augmentation (about 16 000 image-mask pairs), both models improved even more.

The mix of longer training and a larger, more varied dataset gave us the best and most balanced segmentation results.

This shows that both training duration and data diversity are essential for strong learning in infrared image segmentation.

In terms of performance, the U-Net-GAN achieved higher recall and overall segmentation accuracy, especially for small or partially hidden objects.
The ResNet-50-GAN performed well in precision and semantic understanding but needed more computing power and longer training to get there.
Looking at the segmentation masks visually, the U-Net outputs were smoother and more consistent, while the ResNet ones were sometimes blurry or incomplete.
This comparison suggests that matching the architecture to the task is more important than simply using a deeper model.

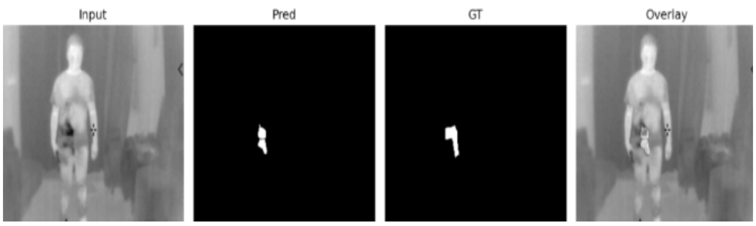
To make the results easier to understand, we created overlay images-combining the predicted mask with the original infrared image.
These overlays clearly show where the detected object appears on the human body and make the results more interpretable for real-world use, like in security or industrial inspection.

Based on this, we also thought about taking the idea one step further-combining thermal (IR) and visible (RGB) images in a single model.
By creating a Visible-Thermal Fusion GAN, we could generate a "fusion image" that shows both body details from the visible image and heat signatures from the infrared one.
This type of fusion could greatly improve detection accuracy and help systems perform better in tough conditions like darkness, fog, or smoke-where normal cameras don't work well.

To sum up, a few main insights came out of this work:

- U-Net work better than deeper residual models for infrared segmentation.
- Giving the model more epochs leads to stable convergence and more realistic masks.
- Strong augmentation helps the model handle variations in thermal data and sensor noise.

In short, the Infrared Segmentation-GAN we developed provides a solid, efficient base for real-time concealed-object detection.
It also opens the door for future work on Visible-Thermal Fusion GANs, which could combine the strengths of both modalities to create more powerful and reliable detection systems.
The following figures shows the relevancy regarding our results and future work.



5. References

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2. 2D-GAN: Dual-Decoding Generative Adversarial Networks for Infrared Image Enhancement- Yang Yu, Lin Jiang, Qijun Hu, Qiang Zeng, Xin Sun, 2024
3. Thermal Image Enhancement using Generative Adversarial Network for Pedestrian Detection- Mohamed Amine Marnissi, Hajer Fradi, Anis Sahbani, Najoua Essoukri Ben Amara, 2023
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7. Deep Residual Learning for Image Recognition- Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun
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9. Automated detection and classification of concealed objects using infrared thermography and convolutional neural networks- WeeLiam Khor, Yichen Kelly Chen, Michael Roberts, Francesco Ciampa, 2024
10. YOLO-IR-Free: An Improved Algorithm for Real-Time Detection of Vehicles in Infrared Images- Zixuan Zhang, Jiong Huang, Gawen Hei, Wei Wang, 2024
11. YOLO-FIRI: Improved YOLOv5 for Infrared Image Object Detection- SHASHA LI, YONGJUN LI, YAO LI, MENGJUN LI, XIAORONG XU, 2021
12. Object tracking and detection techniques under GANN threats: A systematic review- Saeed Matar Al Jaberi, Asma Patel, Ahmed N. AL-Masri, 2023

6. Appendices

Details of choosing the proposed architecture, depends on the following papers' results, materials and summaries:

1. paper[3]

"An infrared camera is a device that forms an image using infrared radiations, compared to the commonly used cameras that form an image using visible light. This camera can detect infrared radiations emitted by an object having a high temperature. Instead of the 400-700 nm range of the visible light camera, infrared one operates in wavelengths as long as 700 nm - 1mm. The main advantage using infrared cameras is that the temperature can be easily measured for dangerous objects while keeping the user out of danger since the thermography is a non-contact method."

"Particularly, we emphasis the need for enhancing the visual quality of images before performing the detection step. To address that, we propose a novel thermal enhancement architecture called TE-GAN based on Generative Adversarial Network, and composed of two modules contrast enhancement and denoising with a post-processing step for edge restoration in order to improve the overall image quality. The effectiveness of the proposed architecture is assessed by means of visual quality metrics and better results are obtained compared to the original thermal images and to the obtained results by other existing enhancement methods."

"Low-resolution together with bad acquisition conditions in some cases, present multiple challenges such as low-contrast, noisy information and blurred details. These challenges make infrared imaging applications hard to perform well. It is essentially the case of various video analysis applications such as object detection, tracking and recognition."
"Generative Adversarial Networks (GANs) are deep learning architectures initially introduced by Goodfellow et al. in [16]. GANs are generally composed of two sub-networks: genera- tive sub-network G and discriminative sub-network D. These architectures have shown excellent performance in image generation and restoration. Also, they have been employed in few studies for image enhancement in visible and thermal domains. For instance, in [17], SRGAN (which refers to Super-Resolution GAN) that includes deep residual network (ResNet) with skip-connection and defines a perceptual loss is proposed. Another related work that made use of GANs for super-resolution infrared image is presented in [18]. Precisely, it employs deep convolutional generative adversarial networks (DCGAN) for infrared face images. Compared to the two previous works that employ GAN architectures for super- resolution in both visible and infrared domains, in [19], the authors rather focused on the problem of enhancing the contrast in infrared images using the conditional generative adversarial networks. Also, in [20], a refined convolutional neural architecture that produces results with higher contrast and sharper details is proposed. But in this approach, visible images are used for training and the network is applied to infrared images."

“a more complete architecture that simultaneously deals with different aspects (low contrast, noise, and blurred edges). Moreover, differently from previous works in thermal domain, where only grayscale converted images are used for training, our proposed architecture is properly trained on impaired images from thermal domain.”
“TE-GAN, which stands for Thermal Enhancement Generative Adversarial Network is shown in Fig.1. The remainder of this section describes each of these architecture components.”
“we propose a new generative adversarial network for thermal image enhancement task. Our proposed architecture is composed of two main modules, the first one is proposed to improve the contrast of the image. Since, by augmenting the contrast, the noise will be gradually more visible, we mitigate this effect by a second module that aims at removing the underlying noise. Both models operate instantly and simultaneously in an end-to-end architecture.”

“YOLOv3 is the third version of object detection al- gorithm in YOLO family. It is one of the most popular and recent real-time object detectors, which improves the accuracy compared to many methods such as ResNet and Feature Pyramid Networks (FPN) structure. The YOLOv3 network makes the prediction at 3 scales to enable multi-scale detection as the case of FPN.”
“It is important to mention that by applying available pre- trained models of YOLOv3 detector trained on visible datasets, using thermal images, a drop in the detection accuracy can be clearly observed, since the two domains (visible and thermal) exhibit different visual characteristics.”
“Once the model is obtained on thermal domain, the detection performance can be compared to the results on the same testing dataset but after enhancing the visual quality of images by our proposed enhancement architecture presented in Section III. Obviously, a prior employment of the enhancement approach can be explored for other applications such as object tracking and recognition in order to improve the overall performance in thermal domain.”
“The baseline detector (YOLOv3) trained and tested on thermal images without enhancement achieves a mAP of 62% and LAMR of 45%. These results are improved while considering the enhancement architecture by a margin of 3% in terms of mAP and 2% in terms of LAMR, which proves the effective- ness of enhancing the image visual quality before detection. By comparing day and night results, as expected, the margin of improvement between with and without enhancement is more significant in nighttime since the temperature decreases during the night.”

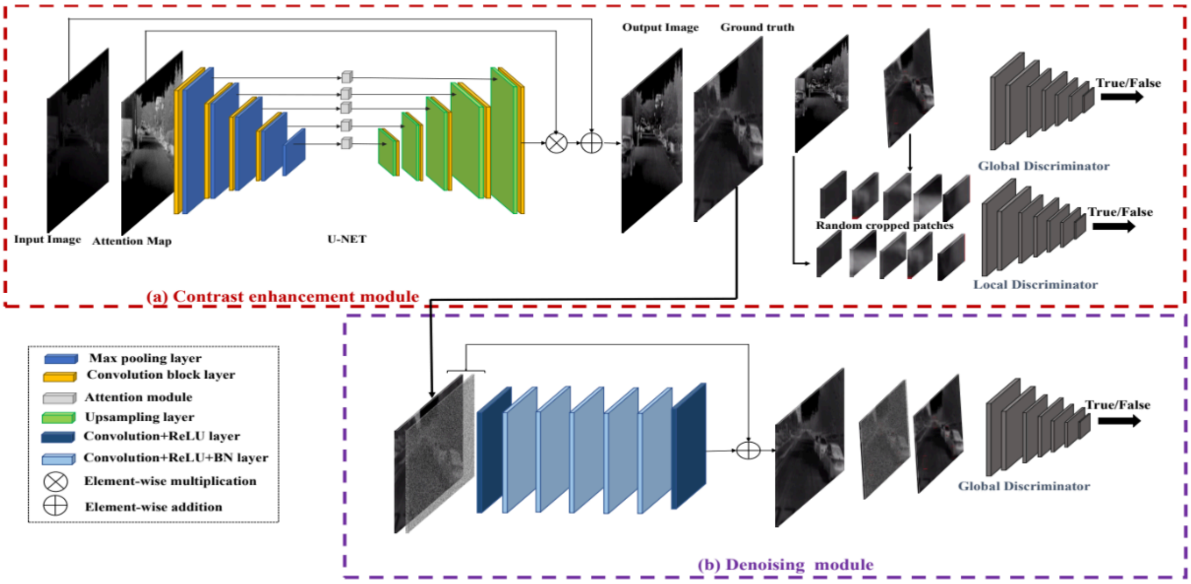
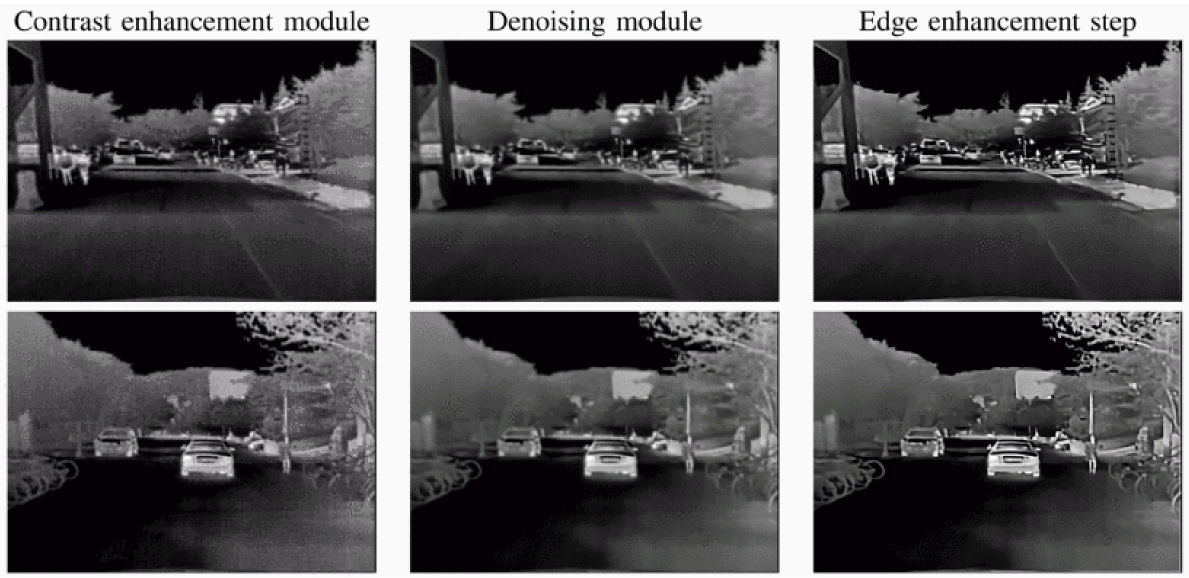


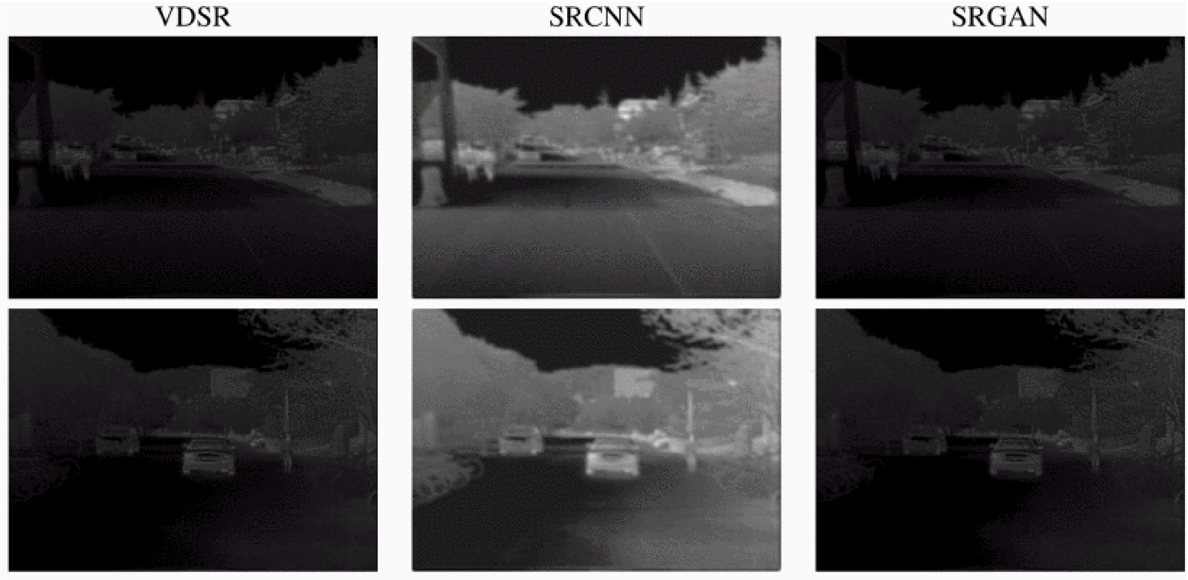
TABLE II
COMPARISON OF THE DETECTION PERFORMANCE OF YOLOV3 DETECTOR
WITH AND WITHOUT ENHANCEMENT

Testing conditions	Metric	Without enhancement	With enhancement
Day	mAP	0.61	**0.63**
	LAMR	0.41	**0.40**
Night	mAP	0.66	**0.73**
	LAMR	0.26	**0.20**
All	mAP	0.62	**0.65**
	LAMR	0.45	**0.43**

Details of intermediate results from the proposed TE-GAN architecture:



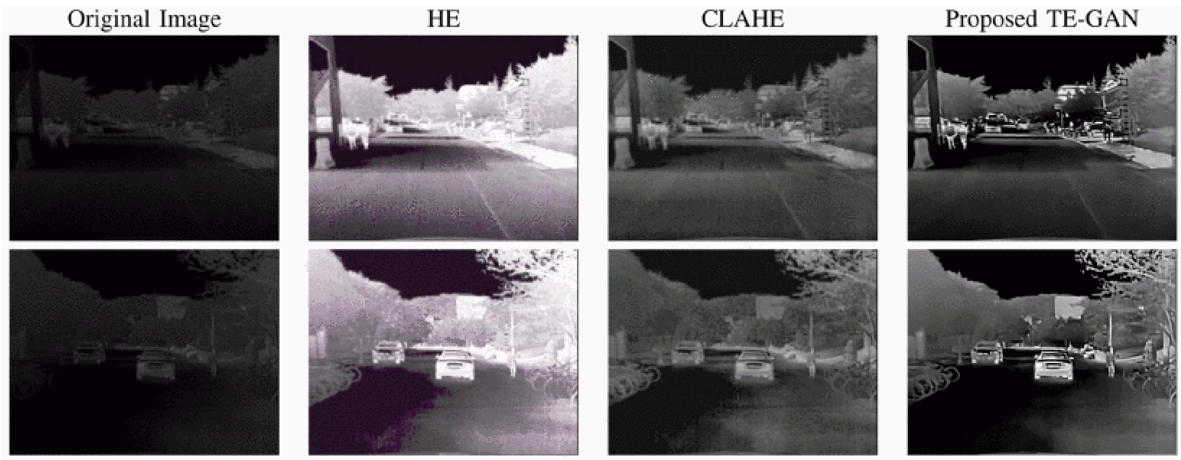
Qualitative results of different super-resolution methods:



Results of the Proposed Te-Gan Architecture:

	HE	CLAHE	TE-GAN
PSNR	7.81	11.92	13.92
SSIM	0.34	0.37	0.50

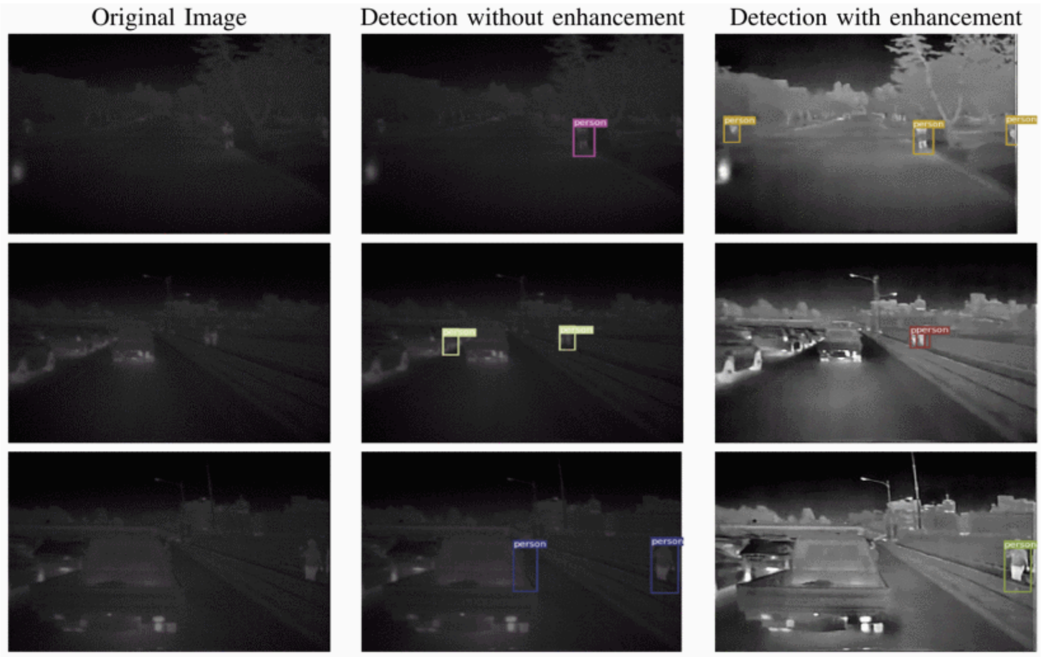
Qualitative results of our proposed architecture TE-GAN:



Results of Pedestrian Detection:

Testing condions	Metri	Without enhancement	With enhancement
Day	mAP	0.61	0.63
	LAMR	0.41	0.40
Night	mAP	0.66	0.73
	LAMR	0.26	0.20
All	mAP	0.62	0.65
	LAMR	0.45	0.43

Some results of pedestrian detection using yolov3 on thermal images from KAIST dataset:

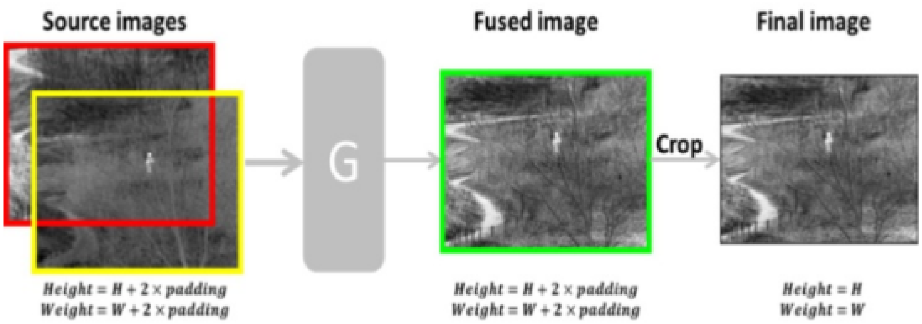


2. paper [5]
Infrared and visible image fusion can obtain combined images with salient hidden objectives and abundant visible details simultaneously. In this paper, we propose a novel method for infrared and visible image fusion with a deep learning framework based on a generative adversarial network (GAN) and a residual network (ResNet). The fusion is accomplished with an adversarial game and directed by the unique loss functions. The generator with residual blocks and skip connections can extract deep features of source image pairs and generate an elementary fused image with infrared
thermal radiation information and visible texture information, and more details in visible images are added to the final images through the discriminator. It is unnecessary to design the activity level measurements and fusion rules manually, which are now implemented automatically. Also, there are no complicated multi-scale transforms in this method, so the computational cost and complexity can be reduced. Experiment results demonstrate that the proposed method eventually gets desirable images, achieving better performance in objective

assessment and visual quality compared with nine representative infrared and visible image fusion methods.

intro:

“ Infrared and visible image fusion, which belongs to multi-modality image fusion, has already been used in image enhancement [3], surveillance [4], remote sensing [5], medical research, etc. The images taken by infrared tensors and visible light tensors focus on different characteristics. Infrared images reflect the thermal radiation of objects [6] and have strong detective capability, making it possible to find targets in bad weather or hidden behind obstructions. However, because of the limits of the imaging principle, infrared images usually have low resolution and few details. Visible images generally have high resolution, evident contrasts, and abundant textures and they are defined in a manner consistent with the human visual system. But it is difficult to get a clear photograph when environmental conditions are not good. Therefore, many researchers have been committed to the subject of infrared and visible image fusion techniques to obtain comprehensive images for further applications.”



Metrics	Formula	Description
EN	$EN = -\sum_{i=0}^{255} p_i \log_2 p_i$	The EN can measure the information contained in the fused image. The larger EN means the better performance.
SF	$SF = \sqrt{(Row\_Freq)^2 + (Column\_Freq)^2}$ $Row_{Freq} = \sqrt{\frac{1}{MN}\sum_{j=0}^{M-1}\sum_{k=1}^{N-1}[F(j,k) - F(j,k-1)]^2}$ $Column_{Freq} = \sqrt{\frac{1}{MN}\sum_{k=0}^{N-1}\sum_{j=1}^{M-1}[F(j,k) - F(j-1,k)]^2}$	The SF indicates the overall activity level in an image, including the row frequency and column frequency. The fused image F will contains more detail features if the value is larger.
SD	$SD = \sqrt{\frac{1}{MN}\sum_{j=1}^{M}\sum_{k=1}^{N}(F(j,k) - \mu)^2}$	μ indicates the mean value of the fused image F. Generally, the larger SD means that F has high contrast and easily attracts visual attention.
MSSIM	$SSIM(X,F) = \frac{(2\mu_X\mu_F + C_1)(2\sigma_{XF} + C_2)}{(\mu_X^2 + \mu_F^2 + C_1)(\sigma_X^2 + \sigma_F^2 + C_2)}$ $MSSIM(A,B,F) = \frac{1}{2M}(\sum_{j=1}^{M} SSIM(A_j,F_j) + \sum_{j=1}^{M} SSIM(B_j,F_j))$	The SSIM index is used to measure the structural similarity between source image (A or B) and fused image F. The final MSSIM uses the mean value to evaluate the overall image quality. μ denotes the mean value of the image. σ represents the standard deviation. $C_1 = (k_1L)^2$ and $C_2 = (k_2L)^2$, k_1 and k_2 are small constants much less than 1. L is the dynamic range of the pixel values (255). A larger SSIM means better fusion process.
CC	$CC = \frac{(r_{AF} + r_{BF})}{2}$ $r_{XF} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(X(i,j) - \overline{X})(\sum_{i=1}^{M}\sum_{j=1}^{N}(F(i,j) - \mu))}{\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N}(X(i,j) - \overline{X})^2(\sum_{i=1}^{M}\sum_{j=1}^{N}(F(i,j) - \mu)^2)}}$	$\overline{X}$ and μ represent the mean value of the source image (A or B) and the fused image F respectively. The larger CC indicates the F gains more information from the source images and the better fusion effect.
VIFF	$VIFF(A,B,F) = \sum_k p_k \cdot VIFF_k(A,B,F)$ $VIFF_k(A,B,F) = \frac{\sum_b FVID_{k,b}(A,B,F)}{\sum_b FVIDN_{k,b}(A,B,F)}$	The VIFF is a multi-resolution image fusion metric using visual information fidelity (VIF) and performs better in human perception matching. The final result is determined by weighting the VIFF of each sub-band. The parameters in the Formula are complex and readers can refer to the original. The VIFF is larger when images have high visual quality.
SCD	$SCD = r_{D_1A} + r_{D_2B}$ $D_1 = F - B$ $D_2 = F - A$ $r_{D_nX} = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N}(D_n(i,j) - \overline{D_n})(\sum_{i=1}^{M}\sum_{j=1}^{N}(X(i,j) - \overline{X}))}{\sqrt{\sum_{i=1}^{M}\sum_{j=1}^{N}(D_n(i,j) - \overline{D_n})^2(\sum_{i=1}^{M}\sum_{j=1}^{N}(X(i,j) - \overline{X})^2)}}$	The SCD is based on the supposition that the difference between one of the input images (A) and the fused image (F) almost discloses the information transferred from the other input image (B). The opposite is also true. The value is larger when fused image contains more information from source images.
FMI	$I(A;F) = \frac{2}{n}\sum_{i=1}^{n}\frac{I_i(A;F)}{H_i(A)+H_i(F)}$ $I(B;F) = \frac{2}{n}\sum_{i=1}^{n}\frac{I_i(B;F)}{H_i(B)+H_i(F)}$ $FMI_F^{AB} = \frac{1}{n}\sum_{i=1}^{n}(\frac{I_i(A;F)}{H_i(A)+H_i(F)} + \frac{I_i(B;F)}{H_i(B)+H_i(F)})$	The FMI calculates the mutual information of the image features using a feature extraction method extracts the feature image of the source (A) and (B) and fused images (F). $I_i(X;F)$ means the regional mutual information and $H_i(X)$ are the entropies. The larger FMI means the better achievement.

Table 4. The average values of the eight metrics on 10 fused images.

Methods	EN	SF	SD	MSSIM	CC	VIFF	SCD	FMI
CVT	6.73408	11.639	29.6285	0.5494	0.5225	0.3693	1.5916	0.3985
DTCWT	6.69775	11.5691	29.2046	0.5579	0.5265	0.3589	1.5957	0.3876
LP	6.84323	11.8508	32.7525	0.5665	0.5214	0.4518	1.612	0.33449
NSCT	6.72252	11.6472	29.6933	0.5761	0.5312	0.4084	1.614	0.38241
TSIFVS	6.83149	11	32.23	0.5749	0.5286	0.4718	1.6278	0.26754
GFF	7.08199	11.1043	40.147	0.5653	0.42	0.2913	1.3317	0.38956
CNN	**7.30549**	11.8958	**48.1136**	0.5673	0.4956	0.5067	1.622	0.32205
Dense-add	6.91024	9.23227	35.66657	**0.59713**	0.55035	0.49293	1.69591	**0.40567**
Dense-L1	6.96276	9.31077	38.90169	0.57984	0.51258	0.3619	1.53546	0.36227
FusionGAN	*7.3033*	*6.4600*	*51.9542*	*0.3352*	*0.7646*	*0.5213*	–	–
Proposed	7.20581	**12.3369**	40.18583	0.57658	**0.56588**	**0.6208**	**1.81625**	0.39641

Our fusion method with GAN and ResNet is a DL-based method that is distinct from traditional ones. Most traditional methods like MSD based, SR based, or others, directly make pixel-level operations on source images and somehow can establish the correlations between the source image and fused image easily. However, our DL-based generative model needs to generate the image through weights and bias of the network which are different to train. The structures of G and D may have a great influence on fusion results. In addition, the loss functions of G and D together determine the optimization strategies and what kinds of information will be added to the final image. So the design of loss functions is another troublesome problem.

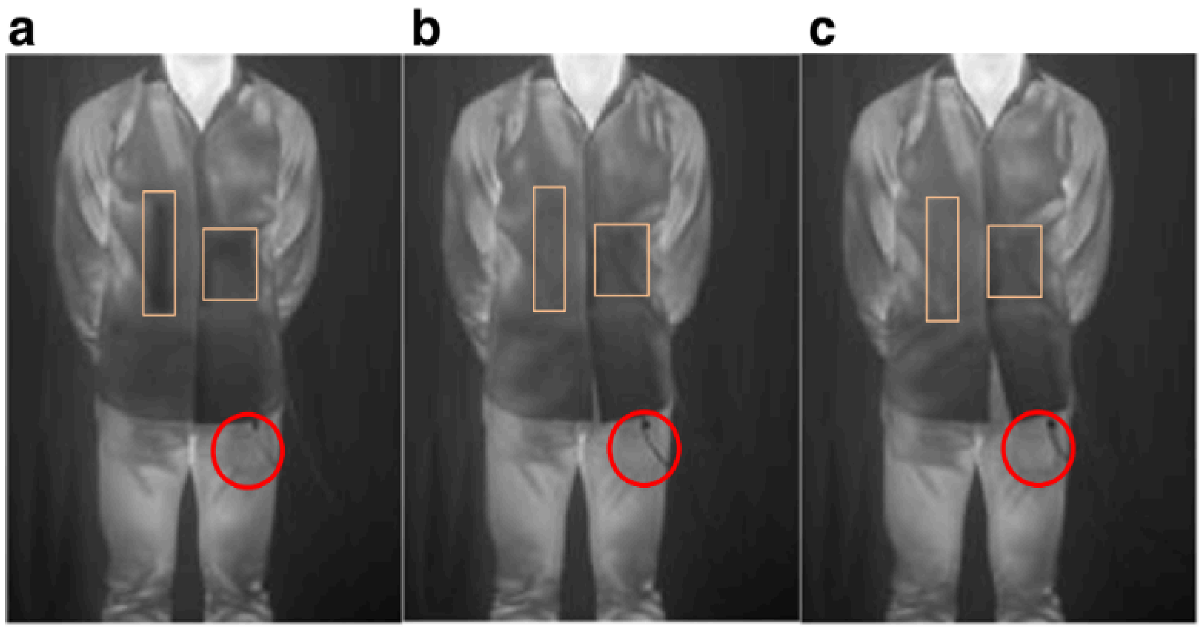
In order to solve these difficulties, firstly, we refer to the mature models of [57,64], and construct the networks of our fusion model with GAN and ResNet. The residual blocks, skip connections, convolution layers, and BN layers work together for feature extraction and combination. The network designed for fusion is other than classification and there is no pooling operation, which may cause the loss of image information. Not only the original residual blocks are introduced, but also the “big residual blocks” are designed. They work together to guarantee that more useful features are generated and retained before reducing the dimensionality to a single-channel fused image. Meanwhile, the ADAM optimizer is adopted with proper settings. Furthermore, the loss function of G in our method consists of adversarial loss and the content loss, which help to increase the intensities of infrared image and the texture details of the visible image. The loss function of D which judges the label image and the image generated by G makes further efforts to add information from visible images.

3. paper[6]

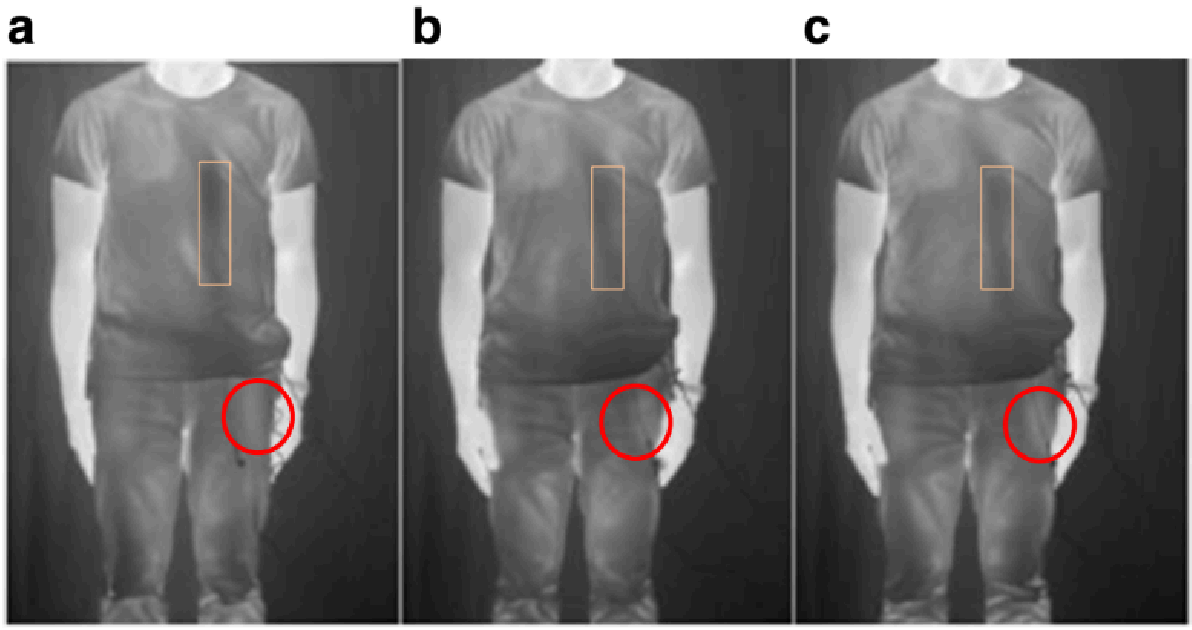
The study presents the comparison of detection and recognition of concealed objects covered with various types of clothing by using passive imagers operating in a terahertz (THz) range at 1.2 mm (250 GHz) and a mid-wavelength infrared (MWIR) at 3-6 μm (50-100 THz). During this study, large dataset of images presenting various items covered with various types of clothing has been collected. The detection and classification algorithms aimed to operate robustly at high processing speed across these two spectrums. Properties of both spectrums, theoretical limitations, performance of imagers and physical properties of fabrics in both spectral domains are described. The paper presents a comparison of two deep learning-based processing methods. The comparison of the original results of various experiments for the two spectrums is presented.

“Non-intrusive detection and recognition of objects concealed under clothes remains challenging in terms of sensors and processing algorithms. ”
“Thermal infrared imagers are able to detect small temperature differences on the object’s surface. ”
“Thermal infrared imagers can be used to detect metallic or non-metallic, potentially dangerous objects hidden under clothes.”
“The thermal infrared imager provides higher spatial resolution but they relies on thermal contrast which decreases due to thermalisation effect.”
“The paper presents a study on passive imaging of concealed objects with automatic detection and classification algorithm.”
“In recent years, machine learning (ML) methods, mostly based on convolutional neural networks (CNN), become increasingly popular. The state-of-the-art CNN architectures include the following: (1) single pass approaches that perform detection within single step (single shot multi-box detector (SSD) [12], you only look once (YOLO) [13]) and (2) region-based approaches that exploit a bounding box proposal mechanism prior to detection (faster regional-CNN (R-CNN) [14], region-based fully convolutional networks (R-FCN) [15], lightweight deep neural networks for real-time object detection (PVANET) [16], local R-CNN [17]).”

“The purpose of this study is to detect and classify potentially dangerous objects including guns and knives placed on the human body covered with the clothing. In the adopted measurement scenario, an item is placed on a human body with constant direct contact. The direct contact results in a heat energy transfer according to Fourier’s law [60, 61].”
“This study concerns passive imaging of concealed objects which relies on radiation emitted by objects. Passive cameras assign temperature differences to differences in the radiated energies in their spectral ranges per unit surface. The detection process relies on searching for temperature differences on the object’s surface. Those differences should correspond to the differences of energy radiated in the whole radiation range. The larger differences may be classified as anomalies and hidden objects. The ability of terahertz and infrared imagers to differentiate temperatures is the main measure for assessing their performance. ”
“Passive Concealed Objects Detection”
“Method for Automatic Object Classification”
YOLO3 and R-FCN(R-FCN is based on the ResNet-101-based [38] framework)

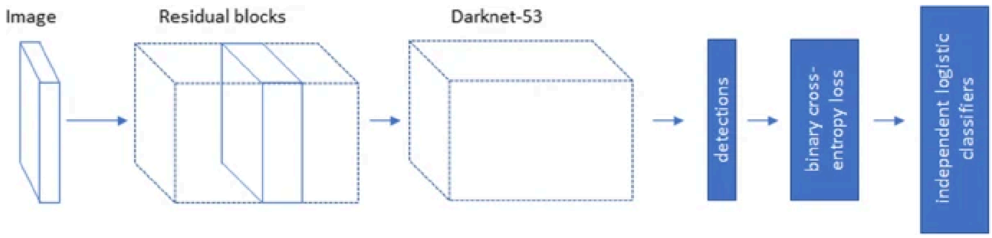


Images presenting a subject wearing a thick cotton shirt, with bomb and a gun hidden under clothes.
MWIR images at the beginning of the measurement (a), after 15 min (b) and after 30 min (c).



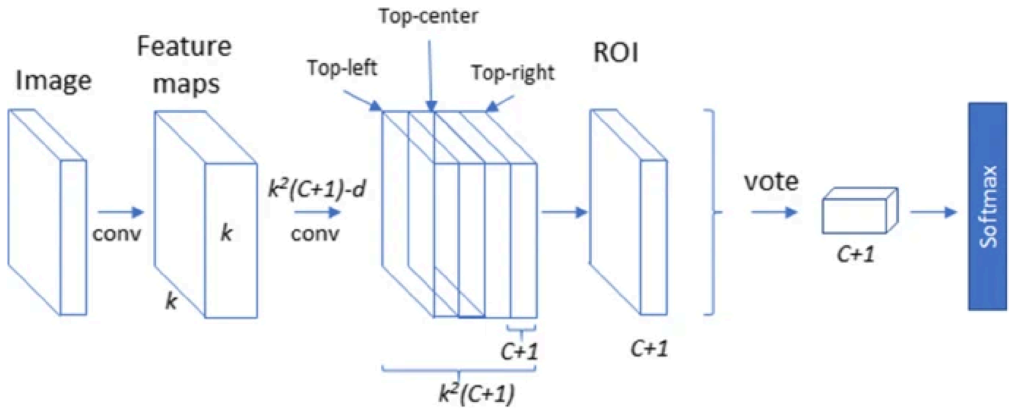
Images presenting a subject wearing a cotton T-shirt, with ceramic knife hidden under clothes. MWIR
images at the beginning of the measurement (a), after 15 min (b) and after 30 min (c).

Fig. 8



Architecture of YOLO3

Fig. 9

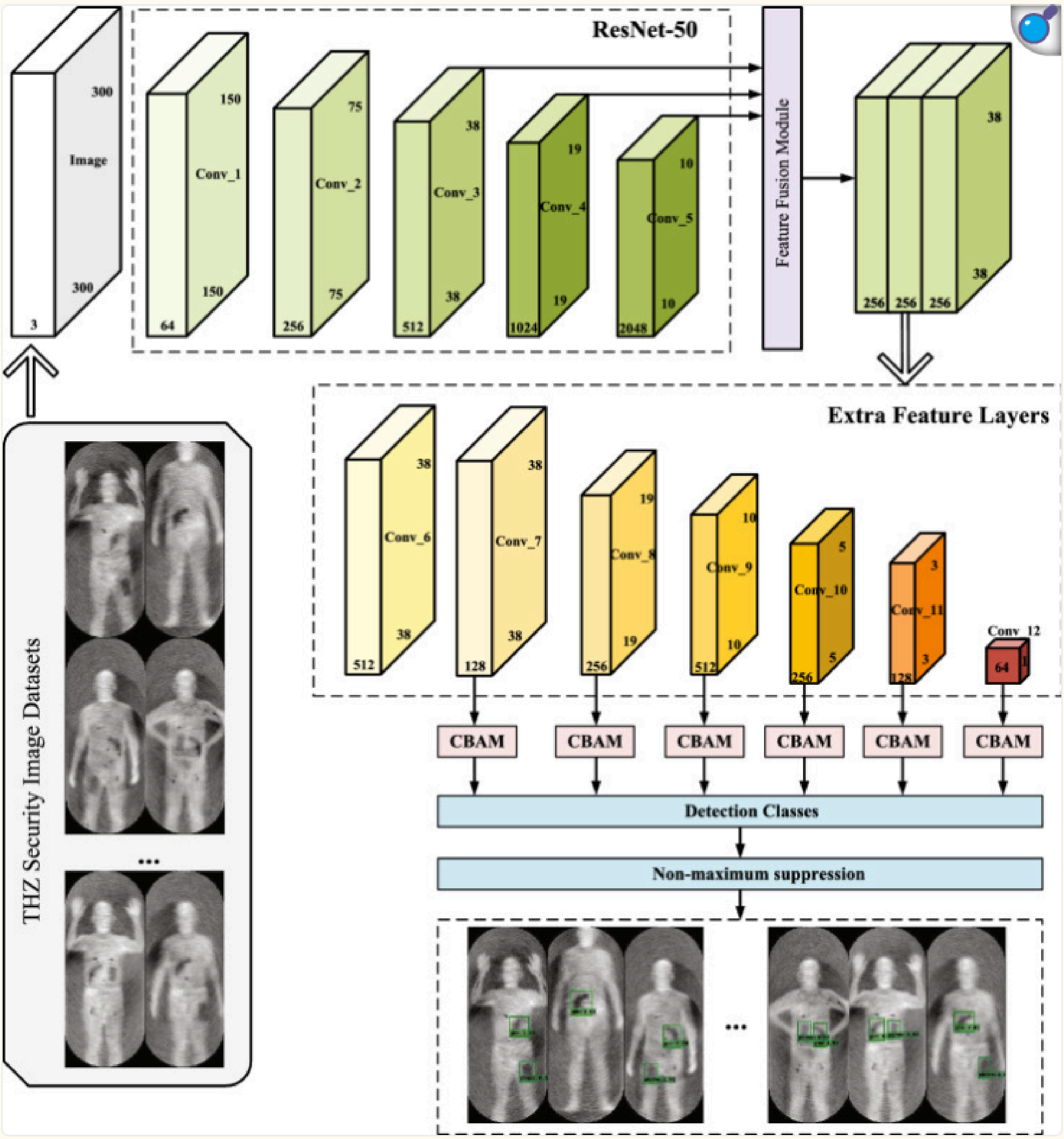


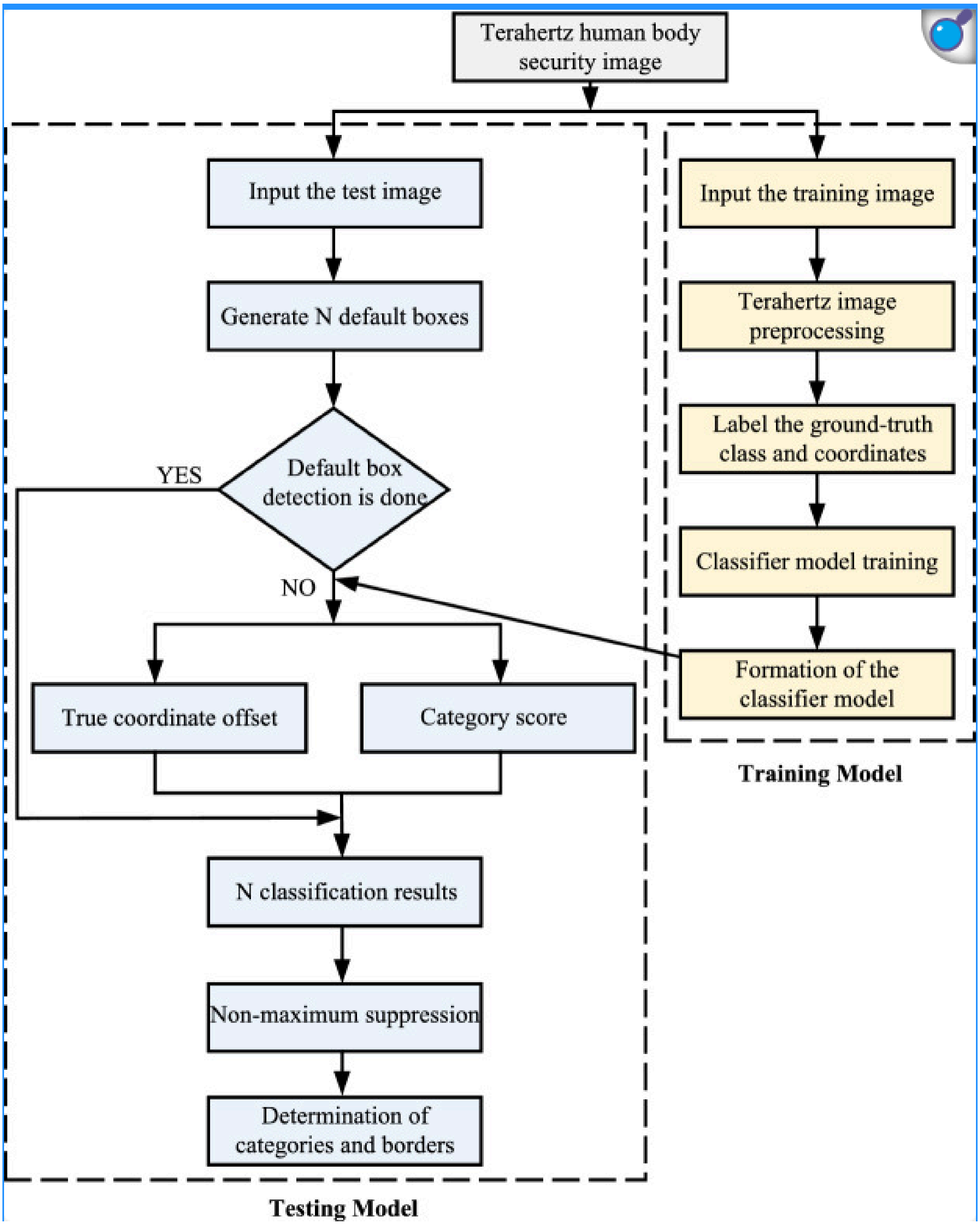
Architecture of R-FCN

3. paper [6]
ResNet-101 vs ResNet-152

4. paper [7]

Figure 2.





5. paper[9]
This paper presents a study on the effectiveness of a convolutional neural network (CNN) in classifying infrared images for security scanning. Infrared thermography was explored as a non-invasive security scanner for stand-of and walk-through concealed object detection. Heat generated by human subjects radiates of the clothing surface, allowing detection by an infrared camera. However, infrared lacks in penetration capability compared to longer

electromagnetic waves, leading to less obvious visuals on the clothing surface. ResNet-50 was used as the CNN model to automate the classifcation process of thermal images. The ImageNet database was used to pre-train the model, which was further fne-tuned using infrared images obtained from experiments. Four image pre-processing approaches were explored, i.e., raw infrared image, subject cropped region-of-interest (ROI) image, K-means, and Fuzzy-c clustered images. All these approaches were evaluated using the receiver operating characteristic curve on an internal holdout set, with an area-under-the-curve of 0.8923, 0.9256, 0.9485, and 0.9669 for the raw image, ROI cropped, K-means, and Fuzzy-c models, respectively. The CNN models trained using various image pre-processing approaches suggest that the
prediction performance can be improved by the removal of non-decision relevant information and the visual highlighting of features.

5. paper[10]
In the field of object detection algorithms, the task of infrared vehicle detection holds significant importance. By utilizing infrared sensors, this approach detects the thermal radiation emitted by vehicles, enabling robust vehicle detection even during nighttime or adverse weather conditions, thus enhancing traffic safety and the efficiency of intelligent driving systems. Current techniques for infrared vehicle detection encounter difficulties in handling low contrast, detecting
small objects, and ensuring real-time performance. In the domain of lightweight object detection algorithms, certain existing methodologies face challenges in effectively balancing detection speed and accuracy for this specific task. In order to address this quandary, this paper presents an improved algorithm, called YOLO-IR-Free, an anchor-free approach based on improved attention mechanism YOLOv7 algorithm for real-time detection of infrared vehicles, to tackle these issues. We introduce a new attention mechanism and network module to effectively capture subtle textures and low-contrast
features in infrared images. The use of an anchor-free detection head instead of an anchor-based detection head is employed to enhance detection speed. Experimental results demonstrate that YOLO-IR-Free outperforms other methods in terms of accuracy, recall rate, and average precision scores, while maintaining good real-time performance.

6. paper[11]
To solve object detection issues in infrared images, such as a low recognition rate and a high false alarm rate caused by long-distance, weak energy, and low resolution, we propose a region-free object detector named YOLO-FIR for the infrared (IR) image with the core of YOLOv5 by compressing channels, optimizing parameters, etc. And an improved infrared image object detection network, YOLO-FIRI, is further developed. Specifically, while designing the feature extraction network, the cross-stage-partial-connections (CSP) module in the shallow layer is expanded and iterated to maximize the use of shallow features. In addition, an improved attention module is introduced in residual blocks to focus on objects and suppress background. Moreover, multiscale detection is added to improve the detection

accuracy of small objects. Experimental results on the KAIST and FLIR datasets show that YOLO-FIRI demonstrates a qualitative improvement compared with the state-of-the-art detectors. Compared with YOLOv4, the mean average precision (mAP50) of YOLO-FIRI is increased by 21% on the KAIST dataset, the speed is reduced
by 62%, the parameters are decreased by 89%, the weight size is reduced by more than 94%, and the computational costs are reduced by 84%. Compared with YOLO-FIR, YOLO-FIRI has an approximately 5% to 20% improvement in AP, AR (average recall), mAP50, F1, mAP50:75, etc. Furthermore, due to the
shortcomings of high noise and weak features, image fusion can be applied to the image preprocessing as a data enhancement method by fusing visible and infrared images based on a convolutional neural network.

7.paper[12]
Infrared imaging technology finds wide application across various fields, yet suffers from issues such as atmospheric thermal radiation interference, resulting in poor contrast, blurred details, and noise in infrared images. Consequently, enhancing infrared images is essential as a preprocessing step to obtain high-quality data. This paper proposes an infrared image enhancement algorithm based on Dual-Decoding Generative Adversarial Networks(2D-GAN) to address these challenges.The proposed algorithm employs a two-step decoding structure within the 2D-GAN framework to enhance the network's capability to comprehend and represent input data effectively. Internal and external skip connections are incorporated to bolster the network's perceptual ability, addressing the loss of detailed information during the encoding and decoding processes. Additionally, a cross-level attention module is designed to dynamically allocate positional weights to feature maps, thereby enhancing the natural appearance of the generated images.The effectiveness of the 2D-GAN-based enhancement algorithm is validated through comparative experiments, supplemented by ablation studies that delineate the contributions of each module within the network. Experimental results demonstrate significant enhancement in infrared image quality, affirming the efficacy of the proposed algorithm.

8. paper[4]
In recent years, deep learning has been used extensively in the field of image fusion. In this article, we propose a new image fusion method by designing a new structure and a new loss function for a deep learning model. Our backbone network is an autoencoder, in which the encoder has a dual branch structure. We input infrared images and visible light images to the encoder to extract detailed information and semantic information respectively. The fusion layer fuses two sets of features to get fused features. The decoder reconstructs the fusion features to obtain the fused image. We design a new loss function to reconstruct the image effectively. Experiments show that our proposed method achieves state-of-the-art performance.

9. paper[1]
The current generative adversarial network (GAN)-based methods for infrared and visible image fusion often overlook global information, leading to inappropriate distribution of fused images in terms of infrared intensity and an overall incongruous presentation. In order to tackle this issue, in this paper, a new global awareness fusion network based on GAN is proposed, termed as GAN-GA. The proposed method comprises a generator and two discriminators. The generator consists of three series-connected global awareness blocks (GABlock), namely detail branch, content branch and feature aggregation block. The detail branch employs a convolutional network and max pooling to extract local detailed information, while the content branch utilizes Transformer to obtain global information. The local details and global information are then passed through the feature aggregation block to focus attention on distinct spatial locations and assign weights based on information importance, resulting in the final fused image. Moreover, the primary and auxiliary concepts are introduced in content loss, incorporating the difference images of infrared and visible as supplementary information into the loss function to fully leverage the information in the source images. For the discriminator, the WGAN-LP loss is employed to constrain its training, which introduces a new gradient penalty based on WGAN-GP to enhance the capability of the discriminator. With these enhancements, GAN-GA effectively captures the global features of the source image while preserving the local texture and infrared intensity, which obtains an overall more coordinated fused image.

10. paper[12]
Current developments in object tracking and detection techniques have directed remarkable improvements in distinguishing attacks and adversaries. Nevertheless, adversarial attacks, intrusions, and manipulation of images/ videos threaten video surveillance systems and other object-tracking
applications. Generative adversarial neural networks (GANNs) are widely used image processing and object detection techniques because of their flexibility in processing large datasets in real-time. GANN training ensures a tamper-proof system, but the plausibility of attacks persists. Therefore, reviewing object tracking and detection techniques under GANN threats is necessary to reveal the challenges and benefits of efficient defence methods against these attacks. This paper aims to systematically review object tracking and detection techniques under threats to GANN-based applications. The selected studies were based on different factors, such as the year of publication, the method implemented in the article, the reliability of the chosen algorithms, and dataset size. Each study is summarised by assigning it to one of the two predefined tasks: applying a GANN or using traditional machine learning (ML) techniques. First, the paper discusses traditional applied techniques in this field. Second, it addresses the challenges and benefits of object detection and tracking. Finally, different existing GANN architectures are covered to justify the need for tamper-proof object tracking systems that can process efficiently in a real-time environment.

Technical details:
https://www.geeksforgeeks.org/generative-adversarial-network-gan/
https://www.geeksforgeeks.org/u-net-architecture-explained/
https://www.geeksforgeeks.org/image-to-image-translation-using-pix2pix/
https://github.com/AmineMarnissi/TE-GAN
https://github.com/vlkniaz/ThermalGAN
https://github.com/hli1221/imagefusion-rfn-nest
https://github.com/eriklindernoren/PyTorch-YOLOv3
https://github.com/jiayi-ma/FusionGAN
https://github.com/RollingPlain/IVIF_ZOO
https://github.com/JorgeFCS/Image-fusion-GAN/tree/main
https://github.com/eyob12/Deep_infrared_and_visible_image_fusion
https://github.com/bupt-ai-cz/LLVIP
https://github.com/hanna-xu/DDcGAN/tree/master/multi-resolution%20vis-ir%20image%20fusion

Datasets:
Concealed Pistol Detection Dataset
https://figshare.com/articles/dataset/Concealed_Pistol_Detection_Dataset/20105600/1?file=35965742
UCLM_Thermal_Imaging_Dataset
https://data.mendeley.com/datasets/b6rpgr6nrh/1
kaist dataset:
https://drive.google.com/file/d/14A3K2IPPPC8-BwPh-YjeHARaZqjnR655/view